

PROJECT :

PROPOSED CONSTRUCTION OF DUTYWA CONFERENCE CENTRE AND ASSOCIATED INFRASTRUCTURE WITHIN THE MBHASHE LOCAL MUNICIPALITY, IN THE EASTERN CAPE.

REPORT:

TERRESTRIAL BIODIVERSITY AND PLANT SPECIES ASSESSMENT REPORT

DATE:

4 August 2026

SPECIALIST:

MBAMBO ECOLOGICAL SERVICES

CLIENT:



**ABANTU
ENVIRONMENTAL
SERVICES**

DEVELOPER:

MBHASHE LOCAL MUNICIPALITY



Mbhashe
Local Municipality
WILLOWVALE | DUTYWA | ELLIOTDALE

DOCUMENT REVISION STATUS

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EXECUTIVE SUMMARY

A full Terrestrial biodiversity Assessment report has been undertaken as part of the Environmental Impact Assessment (EIA) Process for the proposed construction of Dutywa conference centre and associated infrastructure within the Mbhashe Local Municipality, in the Eastern Cape. The DFFE Online Screening Tool identified the area as having Very High Terrestrial Biodiversity Sensitivity because the proposed conference centre occurs within CBA 2 and the Endangered Mthatha Moist grassland. No plant species of conservation concern (SCC's) within the proposed development footprint were identified by the screening tool. The development will include internal roads which will range from 4 to 6 m in width within 12 m road reserves, internal water reticulation from existing mains, with pipeline diameters of 50–110 mm. Stormwater management will involve new 450 mm diameter pipes for certain erven, while others will direct stormwater along roads. Sewer connections will be internal to the layouts, joining the existing adjacent sewer network, with maximum pipe diameters of 160 mm, and short additional connections within existing road reserves where required. The specialist assessment sought to assess the ecological state and current land use of the proposed sites, identify potential sensitive ecosystems and plant and identify potential impacts of the proposed development. A desktop assessment of the site was conducted in terms of current vegetation classifications and biodiversity programmes and plans. Upon completing the desktop assessment, a site visit was conducted on the 28th of July 2026 to assess the actual condition of the terrestrial ecology within the study area.

In accordance with the terms of reference provided, this assessment aimed to:

- Describe the existing vegetation and habitat types within the development footprint,
- Identify the presence or likelihood of Species of Conservation Concern (SCCs),
- Evaluate the site's terrestrial biodiversity sensitivity based on field and desktop assessments,
- Determine the Site Ecological Importance (SEI),
- Assess the significance of potential impacts, and
- Recommend mitigation and management measures to reduce biodiversity loss.

The development footprint is located within the Endangered Mthatha Moist grassland vegetation type. Field surveys confirmed that the area is heavily modified, with significant

disturbance from informal dwellings, domestic animals, widespread alien plant infestations, litter, and scattered remnant thicket patches. No plant species of conservation concern were recorded.

Key Environmental Impacts

- Loss and transformation of secondary grassland
- Loss and reduction of ecosystem functioning and connectivity as a result of vegetation clearance
- Soil and habitat disturbance during construction, including erosion, runoff and alien plant spread, if not properly managed;
- Potential disturbance to nearby wetland;

Key Mitigation Measures

- Minimise the development footprint and restrict vegetation clearance to approved footprint
- Minimise disturbance to the wetland, tributary and associated buffer, and implement appropriate erosion, sediment and stormwater management measures, particularly at the access-road crossing;
- Rehabilitate disturbed areas post-construction, including re-vegetation with locally indigenous grass species;
- Implement an alien vegetation management plan to control alien species, and any new invasions following construction;
- Retain surrounding grassland and ecological linkages as far as practicable.

From a terrestrial biodiversity perspective, the residual impacts of the proposed development are considered manageable provided that the recommended mitigation and rehabilitation measures are strictly implemented.

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LIST OF ACRONYMS

AIPS	ALIEN INVASIVE PLANTS
BGIS	BIODIVERSITY GEOGRAPHIC INFORMATION SYSTEMS
CBA	CRITICAL BIODIVERSITY AREA
CREW	CUSTODIANS OF RARE AND ENDANGERED WILDFLOWERS
DEA	DEPARTMENT OF ENVIRONMENTAL AFFAIRS
DEDEAT	DEPARTMENT OF ECONOMIC DEVELOPMENT, ENVIRONMENTAL AFFAIRS AND TOURISM
DFFE	DEPARTMENT OF FORESTRY, FISHERIES AND THE ENVIRONMENT
EA	ENVIRONMENTAL AUTHORISATION
EAP	ENVIRONMENTAL ASSESSMENT PRACTITIONER
ECBCP	EASTERN CAPE BIODIVERSITY CONSERVATION PLAN
ECPAES	EASTERN CAPE PROTECTED AREA EXPANSION STRATEGY
ECO	ENVIRONMENTAL CONTROL OFFICER
EIA	ENVIRONMENTAL IMPACT ASSESSMENT
EMPR	ENVIRONMENTAL MANAGEMENT PROGRAMME
ESA	ECOLOGICAL SUPPORT AREA
FEPA	FRESHWATER ECOSYSTEM PRIORITY AREA
IAS	INVASIVE ALIEN SPECIES
IUCN	INTERNATIONAL UNION FOR CONSERVATION OF NATURE
NEWPOSA	NATIONAL ENVIRONMENTAL WEB PLANT OCCURRENCE SEARCH APP
NPAES	NATIONAL PROTECTED AREA EXPANSION STRATEGY
PAOI	PROJECT AREA OF INFLUENCE
PoO	PROBABILITY OF OCCURRENCE
SACNASP	SOUTH AFRICAN COUNCIL OF NATURAL SCIENTIFIC PROFESSIONS
SANBI	SOUTH AFRICAN NATIONAL BIODIVERSITY INSTITUTE
SCC	SPECIES OF CONSERVATION CONCERN
SEI	SITE ECOLOGICAL IMPORTANCE

SWSA

STRATEGIC WATER SOURCE AREA

ToPS

THREATENED OR PROTECTED SPECIES

DETAILS OF SPECIALISTS

The details of Specialists are as follows –

The appointed SPECIALIST is registered with the South African Council for Natural Scientific Professions (SACNASP) in terms of Section 18 of the Natural Scientific Professions Act, 2003 (Act No. 27 of 2003), as amended. The details of Specialist are as follows –

Mrs Zona Quvile: zonaquvile@earthguard.co.za

PhD Candidate (Environmental Science)- Rhodes University- Makanda

Main areas of specialisation

- Ecological Reserve Determination (ERD) and Ecological Water Requirement (EWR) studies.
- Wetland delineation, classification, and functional assessment.
- Surface water hydrology and catchment process analysis.
- Assessment of aquatic habitat integrity and ecological condition (PES, EIS).
- Design and implementation of aquatic rehabilitation and restoration plans.
- Preparation of Water Use Licence Applications (WULAs) in terms of Section 21 of the National Water Act (Act 36 of 1998).
- Environmental Impact Assessments (EIAs) for aquatic systems and watercourse-related developments.
- Geographical Information Systems (GIS) mapping and spatial analysis of aquatic ecosystems.
- Specialist consultant for environmental management and compliance monitoring of water-related projects.

Summary of Expertise

Qualifications: Med (Environmental Education), Rhodes University
Biodiversity and Conservation Honours, Rhodes University
BSc Environmental Studies, Walter Sisulu University
Tools for Wetland Assessment Course, Rhodes University
Hydropedology & Wetlands Course (WETREST; Digital Soils Africa)
The mini stream assessment scoring system (miniSASS), GroundTruth

Professional Affiliation: SACNASP; 115598

EAPASA; 2019/1039

Experience with ecological assessments: 10 years

Employment

Jan 2020 – present	Earthguard Consulting, Director, Senior Environmental Scientist
Aug 2017 – Dec 2019	SRK Consulting (Pty) Ltd, Environmental Scientist
Feb 2015 – Jul 2017	Sazi Environmental Consulting, Environmental Consultant
Sep 2014 & Aug 2015	Coastal and Environmental Services, Part-time Junior Environmental Consultant
Mar 2011 – Aug 2011	Walter Sisulu University, Research Assistant

Miss Thandwayo Mbambo

PhD Candidate (Botany) - UKZN

Main areas of specialisation:

- Grassland ecology
- Plant biodiversity and ecological monitoring
- Environmental consulting
- Alien plant identification and control
- GIS and remote sensing

Tertiary Qualifications:

- MSc Botany, UKZN
- BSc Botany (Honours), NMMU
- BSc Environmental Sciences (Botany and Geography), NMMU
-

Registrations and memberships:

- Certificated Natural Scientist (Cert. Sci. Nat) (SACNASP) Registration No. 127029
- Institute of Waste Management of Southern Africa (IWMSA) Reg No. 40121001
- Member of the Grassland Society of Southern Africa Reg No. 2447580

Experience with fauna and flora assessments: 5 years

Employment:

- Mbambo Ecological Services : Feb 2025 – present – Director & Lead Ecologist
- Lwethuma : Jan 2025 – 30 Jan 2025 – Environmental Scientist
- Abantu Environmental Services : Jul 2020 – November 2024 – Environmental Scientist
- SANBI : Oct 2020 – April 2021 – Research Assistant (Alien invasives management)
- GIBB (Pty) Ltd: Oct 2019 – Sep 2020 - Graduate Environmental Scientist
- Nelson Mandela University: Geography & Botany Lecturing Assistant - Jul 2015 – Aug 2018

DECLARATION OF INTEREST

I, Zona Quvile , in my capacity as specialist consultant, hereby declare that I –

- Act as an independent consultant;
- Have expertise in conducting the specialist report relevant to this application, including knowledge of the Act, regulations and any guidelines that have relevance to the proposed activity;
- Do not have any financial interest in the undertaking of the activity, other than remuneration for the work performed in terms of the National Environmental Management Act, 1998 (Act 107 of 1998);
- Have and will not have vested interest in the proposed activity proceeding;
- Have no, and will not engage in, conflicting interests in the proposed activity;
- Undertake to disclose, to the competent authority, any material information that has or may have the potential to influence the decision of the competent authority or the objectivity of any report, plan or document required in terms of the National Environmental Management Act, 1998 (Act 107 of 1998);
- Will provide the competent authority with access to all information at my disposal regarding the application, whether such information is favourable to the applicant or not;
- As a registered member of the South African Council for Natural Scientific Professions, will undertake my profession in accordance with the Code Of Conduct of the Council, as well as any other societies to which I am a member;
- Based on information provided to me by the project proponent and in addition to information obtained during this study, have presented the results and conclusion within the associated document to the best of my professional ability;
- Reserve the right to modify aspects about the present investigation should additional information become available through ongoing research and/or further work in this field; and
- I realise that a false declaration is an offence in terms of Regulation 71 and is punishable in terms of Section 24F of the Act.

Terrestrial Ecologist

04 August 2026

DECLARATION OF INTEREST

I, Thandwayo Mbambo , in my capacity as specialist consultant, hereby declare that I –

- Act as an independent consultant;
- Have expertise in conducting the specialist report relevant to this application, including knowledge of the Act, regulations and any guidelines that have relevance to the proposed activity;
- Do not have any financial interest in the undertaking of the activity, other than remuneration for the work performed in terms of the National Environmental Management Act, 1998 (Act 107 of 1998);
- Have and will not have vested interest in the proposed activity proceeding;
- Have no, and will not engage in, conflicting interests in the proposed activity;
- Undertake to disclose, to the competent authority, any material information that has or may have the potential to influence the decision of the competent authority or the objectivity of any report, plan or document required in terms of the National Environmental Management Act, 1998 (Act 107 of 1998);
- Will provide the competent authority with access to all information at my disposal regarding the application, whether such information is favourable to the applicant or not;
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- Reserve the right to modify aspects about the present investigation should additional information become available through ongoing research and/or further work in this field; and
- I realise that a false declaration is an offence in terms of Regulation 71 and is punishable in terms of Section 24F of the Act.

Terrestrial Ecologist

04 August 2026

1. INTRODUCTION

Mbambo Ecological Services has been appointed by Abantu Environmental Services to conduct a terrestrial biodiversity and plant species assessment for the proposed construction of Dutywa conference centre and associated infrastructure within the Mbhashe local Municipality, in the Eastern Cape. This report follows on from the site verification screening report completed for the project area.

Although the DFFE online screening tool identified the site as having Very High Terrestrial Biodiversity Sensitivity and Medium Plant Species Sensitivity, the actual site sensitivity is considered low due to existing disturbances. As per the assessment protocols identified in the DFFE Screening Report, a plant species and terrestrial biodiversity impact assessment were required. These assessments were conducted following the relevant protocols:

- Protocol for the Specialist Assessment and minimum report content requirements for environmental impacts on Terrestrial Biodiversity (GN 320, published 20 March 2020)
- Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Plant Species (GN 1150 published on 30 October 2020).

The impact assessment methodology of the Species Impact Assessment Guidelines (SANBI 2020) was utilised to determine the impact of the proposed development on floral species.

The objectives of the assessment are to:

- Identify the Environmental Sensitivity of the site using desktop and online resources
- Describe the vegetation types on site, and identified sensitivities
- Determine the threat status and sensitivity of the vegetation type and plant species on site
- Describe the level of degradation of the vegetation and habitats on site
- Assess the impact of the proposed development on the vegetation, inclusive of ecological processes and plant Species of Conservation Concern (SCC), of the site

- Provide recommendations to mitigate the negative environmental impacts on the terrestrial biodiversity of the proposed development

1.1. PROJECT DESCRIPTION

Mbhashe Local Municipality proposes the construction of a conference centre and associated infrastructure in Dutywa.

The proposed development comprises the following:

- Proposed conference centre
- Proposed Sewer Pipe - 160mm Class 34 UPVC and Galvanised Steel Sewer Pipe
- Proposed Water Pipe - main - 8 m servitude
- Proposed Concrete Stormwater Pipes- 450mm – 600mm Ø 100D
- Proposed Sewer Manholes
- Proposed Sw Catchpit/Manhole
- **Proposed concrete/paved/tar access road and a crossing structure (bridge/culvert)**
 - 5960mm (5.96km) length
 - width of 3700mm x 2 lanes (7.4m + 1.65m = 9.5m)
 - shoulders including sidewalks;
 - side walks = 1200mm
 - stormwater drains and should of 465mm
 - **Layers works**
 - 60mm thick 35mpa interlocking brick
 - paving over 20mm compacted sand
 - 150mm g5 material stabilised with 3%
 - cement compacted to 98% mod aashto
 - 150mm g7 material compacted to
 - 95% mo150mm insitu compacted to
 - 93% mod aashtod aashto

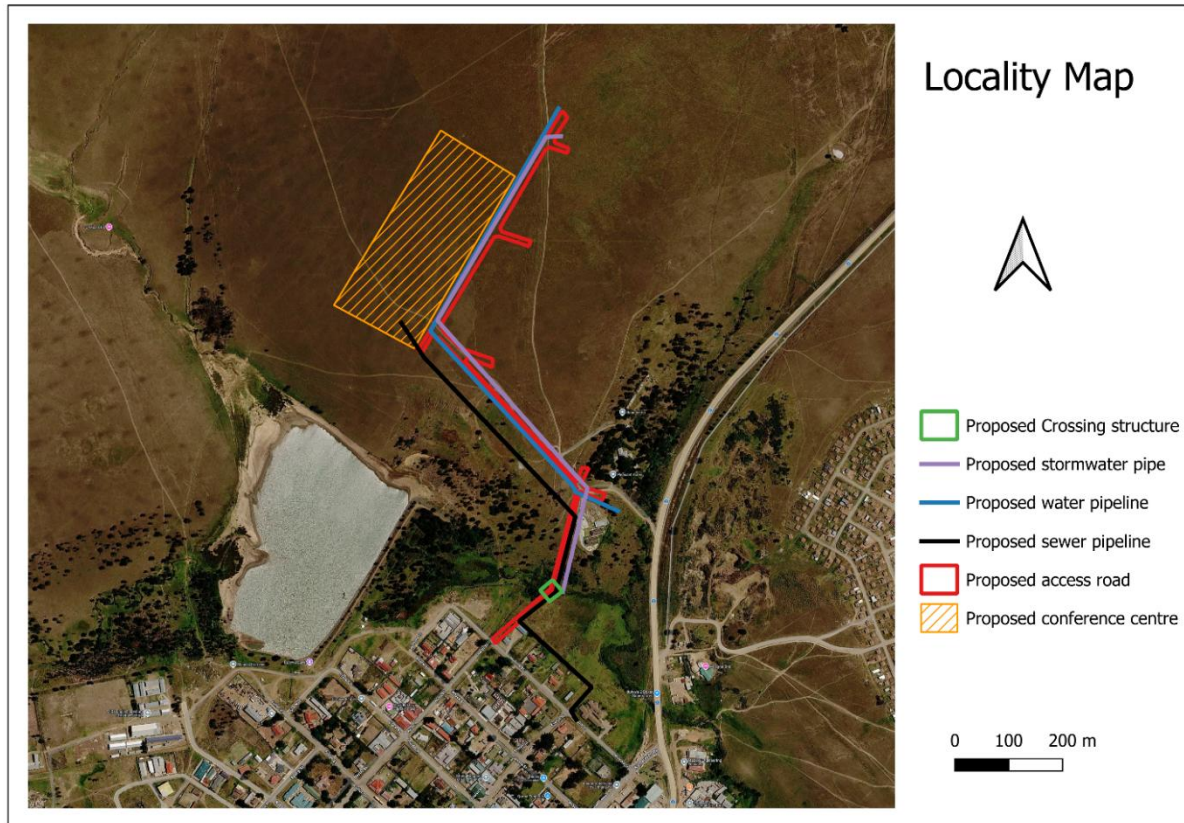


Figure 1: Locality map of the proposed development.

1.2. LAND USE/COVER

The proposed development area is predominantly covered by secondary grassland associated with the Endangered Mthatha Moist Grassland vegetation type. The grass layer is relatively continuous with good vegetation cover. The area is currently utilised for livestock grazing, as evidenced by cattle and sheep/goat dung observed during the site assessment. Weedy and alien invasive plant species are present, particularly in more disturbed areas. Limited litter was observed, mainly along the river crossing and around existing infrastructure. No extensive bare areas or visible soil erosion were observed within the proposed conference-centre footprint.

1.3. ASSUMPTIONS AND LIMITATIONS

The following assumptions and limitations are listed regarding the ecological assessment of the project site:

- The information regarding the proposed development received from the client and EAP is deemed accurate.
- All reasonable measures will be done to compile a species list for the site, including consulting existing species databases and site visit. However, it is not guaranteed that the report will produce a comprehensive species list, as only a single site visit will be done for this assessment.
- The field survey was conducted during the growing season (July 2026), which falls outside the peak flowering period (typically spring to early summer for many Grassland species). As such, the detectability of certain plant species may have been reduced; however, the survey was not intended to record every plant species. Instead, it focused on assessing habitat condition, vegetation structure, ecological functioning, disturbance, and biodiversity sensitivity.

2. LEGAL FRAMEWORK

Environmental Impact Assessments (EIAs) are required in South Africa in terms of the National Environmental Management Act (Act 107 of 1998) (NEMA) and its' associated EIA Regulations (2014) as amended. Developments likely to have a major impact require scoping and EIA, and those likely to have a lesser impact require a Basic Assessment.

If natural vegetation will be affected by a proposed development, a vegetation survey should be commissioned as part of the environmental assessment process. If a subpopulation of a species of conservation concern is found to occur on the proposed development site, it would be one indicator that the proposed activity is likely to result in loss of biodiversity, bearing in mind that loss of subpopulations of these species will either increase their extinction risk or may in fact result in their extinction. The detection of a threatened species on a site during an environmental assessment should result in an Environmental Authorisation from the competent authority that avoids, mitigates, remedies or offsets loss of habitat for the species in question. The competent authority may also refuse authorisation for the proposed activity.

2.1. RELEVANT LEGISLATION

The proposed development will be subject to the requirements of various items of South African legislation (**Table 1**). These are described below.

Table 1. Legislation applicable to the project

Title of legislation or guideline	Application to the project
Section 24 of the Constitution Act (No. 108 of 1996)	States that: Everyone has the right to: • an environment which is not harmful to their health or well-being • have the environment protected for the benefit of present and future generations through reasonable legislative and other measures that: • prevent pollution and ecological degradation • promote conservation • secure ecologically sustainable development and use of natural resources, while promoting justifiable economic and social development. • Obligation to ensure that the proposed development will not result in pollution and ecological degradation; and • Obligation to ensure that the proposed development is ecologically sustainable, while demonstrating economic and social development.
National Environmental Management Act (NEMA) (No. 107 of 1998)	• -The developer must apply the NEMA principles, the fair decision-making and conflict management procedures that are provided for in NEMA; and • -The developer must apply the principles of Integrated Environmental Management and consider, investigate and assess the potential impact of existing and planned activities on the environment, socio-economic conditions and the cultural heritage.
National Environmental Management: Protected Areas Act (No. 57 of 2003)	• The objective of this Act is to provide for the protection and conservation of ecologically viable areas representative of South Africa's biological diversity and its natural landscapes and seascapes; and • In terms of Section 50 (1)(a)(ii) of this Act, the management authority may "Carry out or allow an activity in the reserve aimed at raising revenue". However, Section 50 (2) states that such activity may not negatively affect the survival of any species in, or significantly disrupt the integrity of the ecological system of the nature reserve. Furthermore, in terms Section 51 (a), the Minister or MEC is responsible for the regulations or restrictions of the development and other activities in a protected environment, "which may be inappropriate for the area, given the purpose for which the area was declared".
	•

National Environmental Management Act (NEMA) (107 of 1998; as amended), and the Specialist Assessment Protocols (GNR 320 of 2020)	The contents of this specialist report comply with the legislated requirements as described in the following specialist assessment protocols as listed in the projects' Screening Report: Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Biodiversity.
National Environmental Management: Biodiversity Act (NEMBA), 2004 (Act No. 10 of 2004)	The National Environmental Management: Biodiversity Act (NEMBA), No. 10 of 2004, aims to assist with the management and conservation of South Africa's biological diversity through the use of legislated planning tools. These planning tools include the declaration of bioregions and the associated bioregional plans as well as other mechanisms for managing and conserving biodiversity. The objectives of the Act include inter alia: • The management and conservation of biological diversity within the Republic and of the components of such biological diversity; • The use of indigenous biological resources in a suitable manner; • The fair and equitable sharing of benefits arising from bio-prospecting of genetic material derived from indigenous biological resources; and • To give effect to ratified international agreements relating to biodiversity which are binding on the Republic. • To provide for co-operative governance in biodiversity management and conservation; and • To provide for a South African National Biodiversity Institute to assist in achieving the objectives of the Act. • In addition to this, Sections 50-62 of the Act provide details relating to the protection of threatened or protected ecosystems and species, while Sections 63-77 of the Act provide details relating to alien and invasive species with the purpose of preventing their introduction and spread, managing, controlling and eradicating of alien and invasive species.
National Environmental Management Act (NEMA) Environmental Impact Assessment (EIA) Regulations 2014 as amended (Act No. 107 of 1998)	The activity triggers activities listed in NEMA EIA Regulations GN R. 327 and GN R. 324.
National Water Act, 1998 (Act No. 36 of 1998)	The proposed development must: Conserve endangered ecosystems and protect and promote biodiversity. Assess the impacts of the proposed development on endangered ecosystems. No protected species may be removed or damaged without a permit; and The proposed site must be cleared of alien vegetation using appropriate means.
National Forest Act (84 of 1998)	Requires that a permit be obtained should any forests or protected trees be removed during the construction phase of the project.
Nature and Environmental Conservation Ordinance (No.19 of 1974)	Listing of protected animal species. Permits are required for removal and relocation of any of these listed species in the EC.

2.2. PROTOCOLS REGULATORY REQUIREMENTS

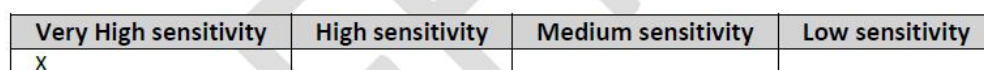
Based on the “Procedures for the Assessment and Minimum Criteria for Reporting on identified Environmental Themes in terms of Section 24(5) (a) and (h) and 44 of the National Environmental Management Act, 1998, when applying for Environmental Authorisation” (“the Protocols”) (Government Notice No. 320 as published in Government Gazette No. 43110 on 20 March 2020, which came into effect on 09 May 2020 and Government Notice No. 1150 as published in Government Gazette No. 43855 on 30 October 2020, which came into effect on the same date), it is hereby noted that the protocols are applicable to the proposed development. According to the Protocols, before commencing with a specialist assessment, the current use of the land and environmental sensitivity of the site under consideration identified by the screening tool must be confirmed by undertaking site sensitivity verification. A site sensitivity verification was conducted on July 2026.

The protocols specified in GN 320 (Protocol for the specialist assessment and minimum report content requirements for environmental impacts on terrestrial biodiversity) of the National Environmental Management Act (NEMA; Act No. 107 of 1998), assessment and reporting requirements for terrestrial biodiversity and plant species are associated with a level of environmental sensitivity identified by the national web-based environmental screening tool (screening tool).

The national web-based Environmental screening tool was queried about the erven of the development.

The terrestrial biodiversity theme indicates that the infrastructure components fall within a **VERY HIGH** area.

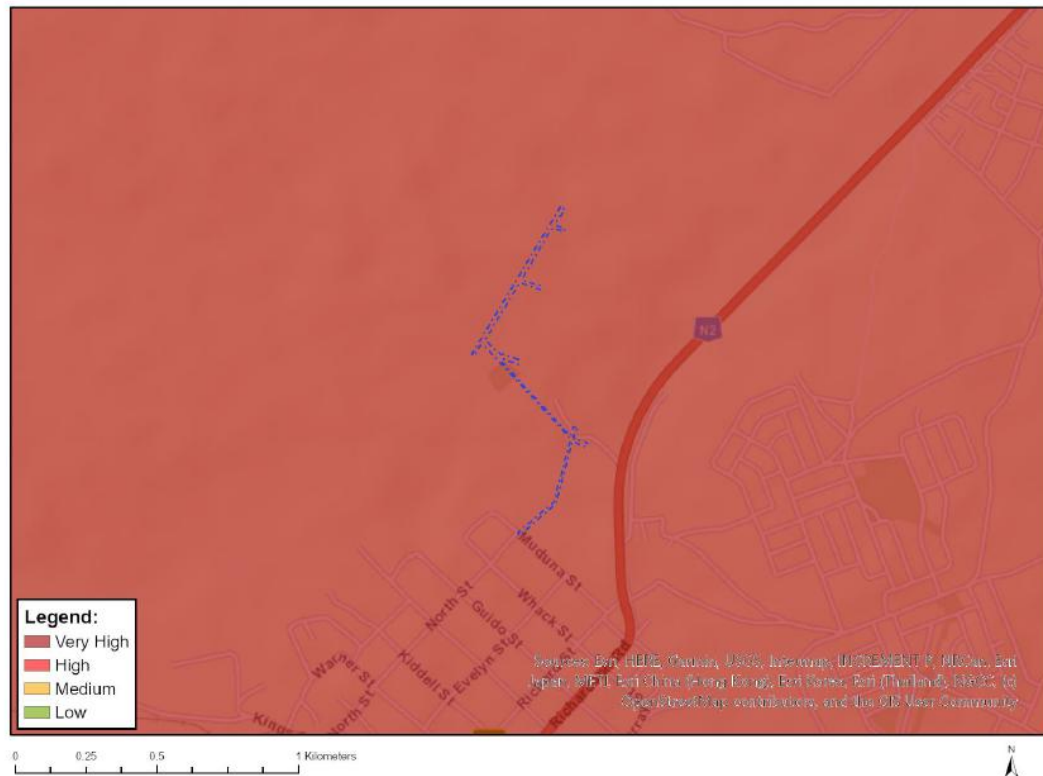
MAP OF RELATIVE TERRESTRIAL BIODIVERSITY THEME SENSITIVITY



Sensitivity	Feature(s)
Very High	CBA2
Very High	EN Mthatha Moist Grassland

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MAP OF RELATIVE TERRESTRIAL BIODIVERSITY THEME SENSITIVITY



Very High sensitivity	High sensitivity	Medium sensitivity	Low sensitivity
X			

Sensitivity Features:

Sensitivity	Feature(s)
Very High	CBA2
Very High	EN Mthatha Moist Grassland

Figure 3: DFFE screening tool extract: terrestrial biodiversity theme for the proposed access road

2.2.2. PLANT SPECIES THEME

No plant SCC's were identified by the screening tool within the proposed development.

MAP OF RELATIVE PLANT SPECIES THEME SENSITIVITY

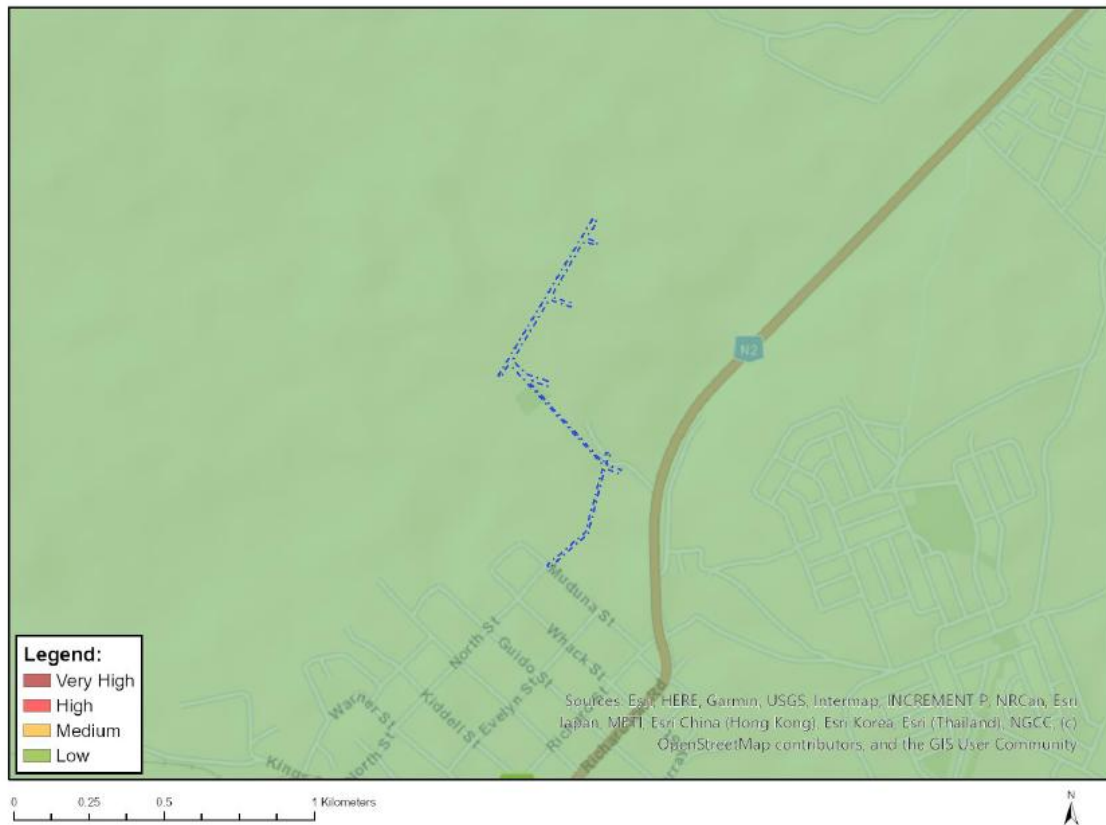


Figure 4: DFFE screening tool extract: plant species theme for the proposed access road

MAP OF RELATIVE PLANT SPECIES THEME SENSITIVITY



Figure 5: DFFE screening tool extract: plant species theme for the proposed conference centre.

2.2. PROJECT ALTERNATIVES

No viable project alternatives were identified or considered for this development. T

2.3. TERMS OF REFERENCE

This specialist study is designed to meet the requirements of the Protocol for the Assessment and Reporting of Impacts on Terrestrial Biodiversity and Plant Species themes.

This site sensitivity assessment follows the requirements of The Environmental Impact Assessment Regulations, as promulgated in terms of Section 24 (5) of the National Environmental Management Act, 1998 (Act No. 107 of 1998), published in GN. No. 320 dated 20 March 2020.

General information

- An applicant intending to undertake an activity identified in the scope of this protocol, on a site identified on the screening tool as being of “**very high sensitivity**” for terrestrial biodiversity, must submit a Terrestrial Biodiversity Specialist Assessment.
- An applicant intending to undertake an activity identified in the scope of this protocol on a site identified by the screening tool as being “**low sensitivity**” for terrestrial biodiversity must submit a Terrestrial Biodiversity Compliance Statement.
- However, where the information gathered from the site sensitivity verification differs from the designation of “very high” terrestrial biodiversity sensitivity on the screening tool and it is found to be of a “low” sensitivity, then a Terrestrial Biodiversity Compliance Statement must be submitted.
- Similarly, where the information gathered from the site sensitivity verification differs from that identified as having a “low” terrestrial biodiversity sensitivity on the screening tool, a Terrestrial Biodiversity Specialist Assessment must be conducted.
- If any part of the proposed development footprint falls within an area of “very high” sensitivity, the assessment and reporting requirements prescribed for the “very high” sensitivity apply to the entire footprint, excluding linear activities for which impacts on terrestrial biodiversity are temporary and the land in the opinion of the terrestrial biodiversity specialist, based on the mitigation and remedial measures, can be returned to the current state within two years of the completion of the construction phase, in which case a compliance statement applies. Development footprint in the context of this protocol means the area on which the proposed development will take place and includes any area that will be disturbed.

Specific outcomes in terms of this report are outlined below:

- To state the indemnity and terms of use of this report as well as to provide the details of the specialists who prepared the report;
- To outline the legislative requirements that were considered for the assessment;
- Compile a desktop assessment with all relevant information as presented by SANBI’s Biodiversity Geographic Information Systems (BGIS) website (<http://bgis.sanbi.org>) and the Environmental Geographical Information Systems (E-GIS) website (<https://egis.environment.gov.za/>). The desktop assessment aims to gain background information on the physical habitat and potential floral and faunal ecology associated with the study area;

- To define the Present Ecological State (PES) of the biodiversity of the study area;
- To determine and describe habitats, communities and the ecological state of the study area;
- To conduct a faunal and floral Species of Conservation Concern (SCC) assessment, including the potential of suitable habitat to occur within the study area for SCC;
- To identify and consider all sensitive landscapes, including rocky ridges, wetlands or any other special features such as Critical Biodiversity Areas (CBAs) and Ecological Support Areas (ESAs);
- To determine the environmental impacts that the proposed development might have on the biodiversity associated with the study area; and
- To develop mitigation and management measures for all phases of the development.

2.4. PURPOSE OF THE BIODIVERSITY ASSESSMENT REPORT

The report content contains the minimum information and reporting requirements for a Terrestrial Biodiversity Impact Assessment and a Terrestrial Plant Species Specialist Assessment, as prescribed in Section 2 of Table 1 of the Protocol for the Terrestrial Biodiversity Theme (GN 320 of 20 March 2020), and is summarised in Table 2 & Table 3 below.

Table 2: Minimum information and report requirements for a terrestrial biodiversity impact assessment report.

Prescribed content of a Biodiversity Assessment Report	Reference in this report
(a) contact details of the specialist, their SACNASP registration number, their field of expertise and a curriculum vitae.	Before Section 1
(b) a signed statement of independence by the specialist.	Before Section 1
(c) a statement on the duration, date and season of the site inspection and the relevance of the season to the outcome of the assessment	Section 3
(d) a description of the methodology used to undertake the site verification and impact assessment and site inspection, including equipment and modelling used, where relevant;	Section 3
(e) a description of the assumptions made and any uncertainties or gaps in knowledge or data as well as a statement of the timing and intensity of site inspection observations;	Section 1
(f) a location of the areas not suitable for development, which are to be avoided during construction and operation (where relevant);	Section 5

Prescribed content of a Biodiversity Assessment Report	Reference in this report
(g) additional environmental impacts expected from the proposed development;	Section 5
(h) any direct, indirect and cumulative impacts of the proposed development;	Section 5
(i) the degree to which impacts and risks can be mitigated;	Section 5
(j) the degree to which the impacts and risks can be reversed;	Section 5
(k) the degree to which the impacts and risks can cause loss of irreplaceable resources;	Section 5
(l) proposed impact management actions and impact management outcomes proposed by the specialist for inclusion in the Environmental Management Programme (EMPr);	Section 5
(m) a motivation must be provided if there were development footprints identified as per paragraph 2.3.6 above that were identified as having a “low” terrestrial biodiversity sensitivity and that were not considered appropriate;	Not applicable
(n) a substantiated statement, based on the findings of the specialist assessment, regarding the acceptability, or not, of the proposed development, if it should receive approval or not; and	Section 6
(o) any conditions to which this statement is subjected.	Section 6
(p) The findings of the Terrestrial Biodiversity Specialist Assessment must be incorporated into the Basic Assessment Report or the Environmental Impact Assessment Report, including the mitigation and monitoring measures as identified, which must be incorporated into the EMPr where relevant.	Section 6
(q) A signed copy of the assessment must be appended to the Basic Assessment Report or Environmental Impact Assessment Report.	Noted

Table 3: Summary of Terrestrial Plant Species Specialist Assessment Requirements.

Requirement	Section in this report
4.1. Medium sensitivity data represents suspected habitat for SCC based on occurrence records for these species collected prior to 2002 and/or is based on habitat suitability modelling.	2.2.2
4.2. The presence or likely presence of the SCC identified by the screening tool, must be confirmed through a site inspection by a specialist registered with the SACNASP in a field of practice relevant to the taxonomic group (“taxa”) for which the assessment is being undertaken.	2.2
4.3. The assessment must be undertaken within the study area.	2.7
4.4. The site inspection to determine the presence or likely presence of SCC must be undertaken in accordance with the <i>Species Environmental Assessment Guideline</i> ¹⁵ .	2.7

4.5. The site inspection is to confirm the presence, likely presence or confirmed absence of a SCC within the site identified as “medium” sensitivity by the screening tool.	2.7
4.6. Where SCC are found on site or have been confirmed to be likely present, a Terrestrial Plant Species Specialist Assessment must be submitted in accordance with the requirements specified for “very high” and “high” sensitivity in this protocol.	3.7.2
4.7. Similarly, where no SCC are found on site during the investigation or if the presence is confirmed to be unlikely, a Terrestrial Plant Species Compliance Statement must be submitted.	

3. METHODS

The terrestrial biodiversity assessment involved a desktop literature survey, as well as a site assessment that took place on 28 July 2026. A comprehensive observed plant species list was produced and annotated according to the relevant legislation. All Threatened or Protected Species were identified, as well as any Invasive Alien Species (IAS).

The approach used in this terrestrial biodiversity assessment, inclusive of terrestrial plant species, is as follows:

3.1. PROJECT AREA OF INFLUENCE

The Project Area of Influence is defined by the important ecosystem processes and functions that may be affected by the proposed development and its activities. The Species Environmental Assessment Guideline (2020) requires that the EAP and Specialists define the taxon-specific Project Area of Influence (PAOI) based on the spatial location of the project (footprint) and the potential extent of the impacts of the anticipated project activities.

3.2. DESKTOP ASSESSMENT

A desktop assessment of the potential plant species, vegetation types and sensitivities of the site based on data extracted from:

- Mucina and Rutherford’s (2009) vegetation map and 2018 updated vegetation map and vegetation descriptions;
- National Environmental Management: Biodiversity Act (Act No. 10 of 2004): Revised National List of Threatened Ecosystems (2022);
- SANBI 2018 final Vegetation map of South Africa, Lesotho and Swaziland

- The IUCN red list category for plant species, as well as supplementary information on habitats and distribution, was obtained from the SANBI threatened species programme (Red List of South African plants, <https://redlist.sanbi.org>)
- Eastern Cape Biodiversity Conservation Plan (ECBCP, 2019); and

3.3. SITE ASSESSMENT

A site sensitivity verification (SSV) was done to determine whether an impact assessment or compliance statement will be necessary to meet the requirements of the protocols for the terrestrial biodiversity and plant species screening report themes.

The footprint of the proposed construction of Dutywa conference centre and associated infrastructure within the Mbhashe local municipality, in the Eastern Cape was assessed. The sites fall within the Grassland biome, which are characterised by summer rainfall, with peak precipitation and flowering typically occurring between October and March. Site sensitivity verification entailed a combination of desktop assessment and field surveys. This method involved walking through the proposed development footprint and recording all plant species and the condition of their habitats on site.

The desktop screening did not identify any Species of Conservation Concern (SCC) to occur in the area. Furthermore, no individuals of SCC were observed during the site visit.

The site visit was considered representative and adequate for an initial assessment of botanical diversity and species composition. Specific features of potential concern were investigated in the field, including the following:

- General vegetation status, i.e. whether the vegetation was natural, disturbed/secondary or transformed;
- Presence of habitats of conservation concern in terms of high biodiversity, presence of species of conservation concern, specific sensitivities (e.g. wetlands), and other indicators of elevated biodiversity or ecological value not evident from the desktop assessment;
- Presence of protected trees; and
- Potential presence of SCC, based on direct observation or presence of suitable habitats for any of the species identified through the desktop screening.

3.4. SITE ECOLOGICAL IMPORTANCE

The species environmental assessment guidelines require that a Site Ecological Importance (SEI) be calculated to determine the importance of a project area for species spatially (SANBI, 2020a). the SEI is considered to be a function of the biodiversity importance (BI) of the receptor (e.g. species of conservation concern, the vegetation/fauna community or habitat type present on the site²⁰) and its resilience to impacts (receptor resilience [RR]).

The detailed methodology is available in the Species Environmental Assessment Guidelines (SANBI, 2020a).

Guidelines for development activities within different importance levels are given in the Table 4 below:

Table 4: Guidelines for interpreting SEI in the context of the proposed development activities.

Site ecological importance	Interpretation to proposed development activities
Very High	Avoidance mitigation – no destructive development activities should be considered. Offset mitigation not acceptable/ not possible (i.e. last remaining populations of species, last remaining good condition patches of ecosystems/ unique species assemblages). Destructive impacts for species/ ecosystems where persistence target remains.
High	Avoidance mitigation wherever possible. Minimisation mitigation – changes to project infrastructure design to limit the amount of habitat impacted; limited development activities of low impact are acceptable. Offset mitigation may be required for high-impact activities.
Medium	Minimisation and restoration mitigation – development activities of medium acceptable, followed by appropriate restoration activities.
Low	Minimisation and restoration mitigation - development activities of medium to high acceptable, followed by appropriate restoration activities.
Very low	Minimisation mitigation - development activities of medium to high acceptable, and restoration activities may not be required.

3.5. IMPACT ASSESSMENT METHODOLOGY

Impacts identified were assessed according to the criteria outlined below. Each impact was ranked according to the nature, extent, duration, magnitude, probability, irreplaceable loss of resources and reversibility (Table 5). These criteria are based on the Department of Environmental Affairs and Tourism (DEAT) Guideline Document to the EIA Regulations (1998). A significance rating was calculated as per the methodology outlined below.

The significance rating of each identified impact / effect was further reviewed by the EAP and/or specialist by applying professional judgement. Where possible, mitigatory measures were recommended for the impacts identified.

Table 5. Impact Rating Methodology

Status of impact	
<i>The environmental impacts of an activity are those resultant changes in environmental parameters, in space and time, compared with what would have happened had the activity not been undertaken. It is an appraisal of the type of effect the activity would have on the affected environmental parameter. Its description includes what is being affected, and how</i>	
Indication whether the impact is adverse (negative) or beneficial (positive).	+ ve (positive – a ‘benefit’)
	– ve (negative – a ‘cost’)

EXTENT OF THE IMPACT		
This addresses the physical and spatial scale of the impact.	Site – The impact area extends only as far as the activity – i.e. within the boundaries of the development site.	1
	Local - The impacted area extends slightly further than the actual physical disturbance footprint and could affect the whole, or a measurable portion of adjacent areas (within approx. 5 km of the development site).	2
	Landscape - The impact could affect all areas generally visible to the naked eye, as well as those areas essentially linked to the site in terms of ecosystem functioning	3
	Regional - The impact could affect the site including the neighbouring areas, transport routes and surrounding towns etc.	4
	Ecosystem - The impact could affect areas essentially linked to the site in terms of significantly impacting ecosystem functioning.	5
	National - The impacted area extends beyond provincial boundaries.	6
	International - The impacted area extends beyond national boundaries.	7
DURATION OF IMPACT		
This describes the predicted lifetime / temporal scale of the predicted impact.	Short term - Quickly reversible. Less than the project lifespan. The impact will either disappear with mitigation or will be mitigated through natural process in a span shorter than any of the project phases or within 0 -5 years.	1
	Medium term – Some mitigation will be required to reduce the duration of the impact – 6-15 years.	3
	Long term - the impact will cease when the operation stops.	5
	Permanent - no mitigation measure will reduce the impact after construction.	7

MAGNITUDE OF THE IMPACT		
This provides a qualitative assessment of the severity of a predicted impact / effect.	None - where the aspect will have no impact on the environment	0
	Minor - The affected environment is altered, but natural function and process continue.	1

	Low - where the impact affects the environment in such a way that the natural, cultural and social functions / processes are slightly affected.	2
	Moderate - where the affected environment is altered but natural, cultural and social functions / processes continue, albeit in a modified way	3
	High - natural, cultural or social functions / processes are altered to the extent that they will temporarily cease.	4
	Very High - natural, cultural or social functions / processes are altered to the extent that they will permanently cease.	5
IRREPLACEABLE LOSS OF RESOURCES		
Environmental resources cannot always be replaced; once destroyed, some may be lost forever. It may be possible to replace, compensate for or reconstruct a lost resource in some cases, but substitutions are rarely ideal. The loss of a resource may become more serious later, and the assessment must take this into account.	Short-term – Quickly recoverable. Less than the project lifespan. The resource can be renewed / recovered with mitigation or will be mitigated through natural process in a span shorter than any of the project phases, or in a time span of 0 to 5 years.	1
	Loss of an 'expendable' resource - one that is not deemed critical for biodiversity targets, planning goals, community welfare, agricultural production, or other criteria.	2
	Medium term – The resource can be recovered within the lifespan of the project. The resource can be renewed / recovered with mitigation or will be mitigated through natural process in a span between 5 and 15 years.	3
	Loss of an 'at risk' resource - one that is not deemed critical for biodiversity targets, planning goals, community welfare, agricultural production, or other criteria, but cumulative effects may render such loss as significant.	4
	Long term – The loss of a non-renewable / threatened resource which cannot be renewed / recovered with, or through, natural process in a time span of over 15 years, but can be mitigated by other means.	5
	Permanent – The loss of a non-renewable / threatened resource which cannot be renewed / recovered with, or through, natural process in a time span of over 15 years, or by artificial means.	7

REVERSIBILITY / POTENTIAL FOR REHABILITATION		
The distinction between reversible and irreversible impacts is a very important one and the irreversible impacts not susceptible to mitigation can constitute significant impacts in an EIA (Glasson <i>et al</i> , 1999). The potential for rehabilitation is the major determinant factor when considering the temporal scale of most predicted impacts.	Short term – The impact / effect will be returned to its benchmark state through mitigation or natural processes in a span shorter than any of the phases of the project, or in a time span of 0 to 5 years.	1
	Medium term – The impact / effect will be returned to its benchmark state through mitigation or natural processes in a span shorter than the lifetime of the project, or in a time span between 5 and 15 years.	3
	Long term - The impact / effect will be returned to its benchmark state through extensive mitigation or natural processes in a time span between 15 and 25 years.	5
	Permanent – The impact/ effect is permanent and will never be returned to is benchmark state	7
PROBABILITY OF OCCURRENCE		
The likelihood of the impact actually occurring.	Remote possibility / unlikely	0
	Possibility	1
	Low probability / anticipated	2
	Medium probability / strongly anticipated	3

	High probability / to be expected	4
	Absolute certainty / will occur	5
IMPACT SIGNIFICANCE		
The overall significance of an impact / effect has been ascertained by attributing numerical ratings to each identified impact. The numerical scores obtained for each identified impact have been multiplied by the probability of the impact occurring before and after mitigation. High values suggest that a predicted impact / effect is more significant, whilst low values suggest that a predicted impact / effect is less significant.		
((Spatial Extent + Magnitude + Duration) * Probability) = Significance.		
	Overall Score	
Insignificant – the impact is meaningless has no influence on the decision to develop	< 15	
Low – the impact would not have a direct influence on the decision to develop in the area;	16 - 35	
Medium – the impact could influence the decision to develop in the area unless it is effectively managed / mitigated; and	36 - 65	
High - the impact must have an influence on the decision-making process for development in the area.	> 65	
MITIGATION		
In terms of the assessment process the potential to mitigate the negative impacts is determined and rated for each identified impact and mitigation objectives that would result in a measurable reduction or enhancement of the impact are considered. The significance of environmental impacts has therefore been assessed considering any proposed mitigation measures. The significance of the impact “without mitigation” is therefore the prime determinant of the nature and degree of mitigation required.		

Confidence of assessment	
The degree of confidence in predictions based on available information, Specialist’s judgment and/or specialist knowledge.	Low
	Medium
	High

Impact Mitigation Hierarchy and Residual Impacts

A residual biodiversity impact is the impact of an activity, or activities, on biodiversity that remains after all efforts have been made to avoid and minimise the impacts of the activity, or activities, and to rehabilitate the affected area to the fullest extent possible (DFFE, 2023).

After specialists and EAPs, as part of an EIA, identify and assess all negative biodiversity impacts of a development, they must investigate alternative project locations, designs, technologies, scales and layouts to determine if and how potentially significant negative

impacts on biodiversity could be avoided or minimised, and how effective rehabilitation practices can be.

If the EIA determines that all negative impacts on biodiversity cannot be avoided, and/or that impact minimisation and rehabilitation of the affected area cannot fully mitigate the impacts of the activity/ies on biodiversity, the proposed development would have residual negative biodiversity impacts. The mitigation hierarchy, as illustrated in Figure 5, should be followed to determine if there will likely be residual impacts.

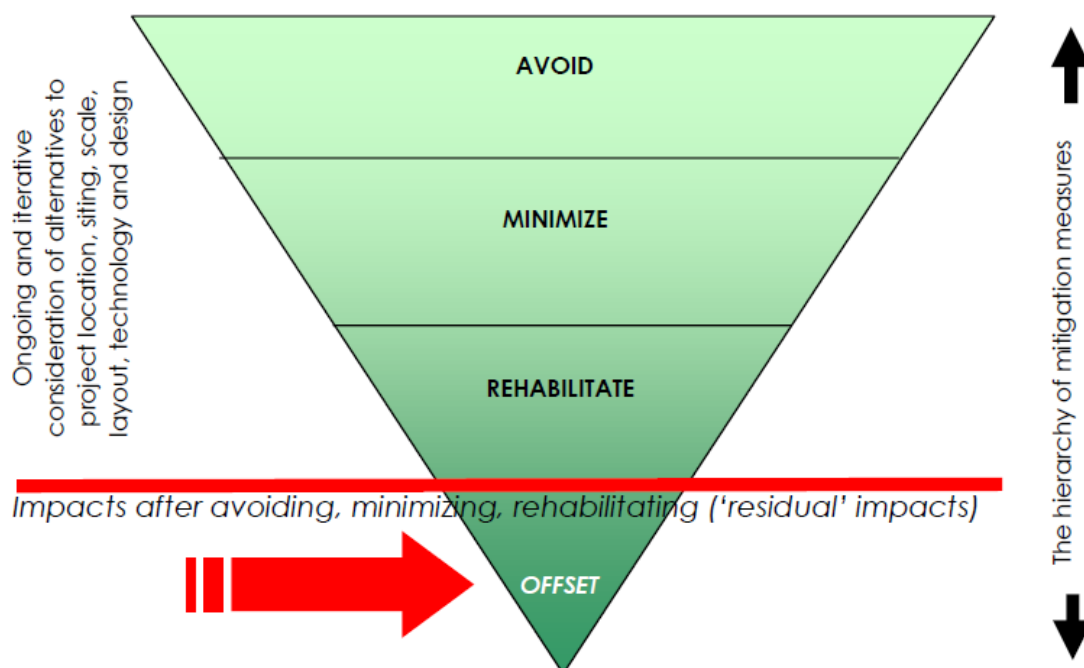


Figure 5: Impact mitigation hierarchy (DFFE, 2023).

Where residual negative biodiversity impacts are evaluated to be of medium or high significance, a biodiversity offset would be required. Impacts of low significance should not require an offset, whereas biodiversity offsets are not deemed appropriated for impacts of very high significance. Very high residual negative impacts that result in the loss or irreplaceable biodiversity should result in the development being considered fatally flawed and avoided

4. STUDY SITE DESCRIPTION

4.1. VEGETATION TYPE

4.1.1. NATIONAL VEGETATION ASSESSMENT

The Vegetation Map for South Africa, Lesotho and Swaziland (VegMap) by Mucina & Rutherford 2006 (Mucina and Rutherford, 2006) is the most widely accepted classification of South Africa's vegetation. It includes information on the conservation status and indicator species for each recognised vegetation type in the country. This biodiversity planning product also forms the basis for the NEM: Biodiversity Act list of Threatened Ecosystems. The project area falls in the Mthatha Moist Grassland as described by Mucina and Rutherford (2006) in the Grassland biome. The Mthatha Moist Grassland is listed as Endangered (South African National Biodiversity Institute, 2021). The grassland biome in South Africa has a high biodiversity, second only to the Fynbos (SANBI, 2013). The vegetation types described below are from (Mucina et al., 2006) and the dominant species are listed in (Figure 6; Table 6).

Mthatha Moist Grassland

Distribution

Eastern Cape Province: Plains between Mthatha and Butterworth parallel to the coastline and excluding the 600–1 080 m.

Vegetation and landscape features

Undulating plains and hills supporting species-poor, sour, wiry grassland with *Eragrostis plana* and *Sporobolus africanus*; although in good condition, it is more likely to be dominated by *Themeda triandra*.

Geology and soils

Mudstones of the Tarkastad and Adelaide Subgroups (Beaufort Group, Karoo Supergroup) underlie this area, with highly leached soils typical of the Fa land type.

Climate

Summer rainfall, with MAP 600–970 mm. The coefficient of variation of MAP 25–30% across the unit but drops to approximately 15% on the southeast-facing inland mountain slopes. Incidence of frost is 2–14 days but is higher further from the coast.

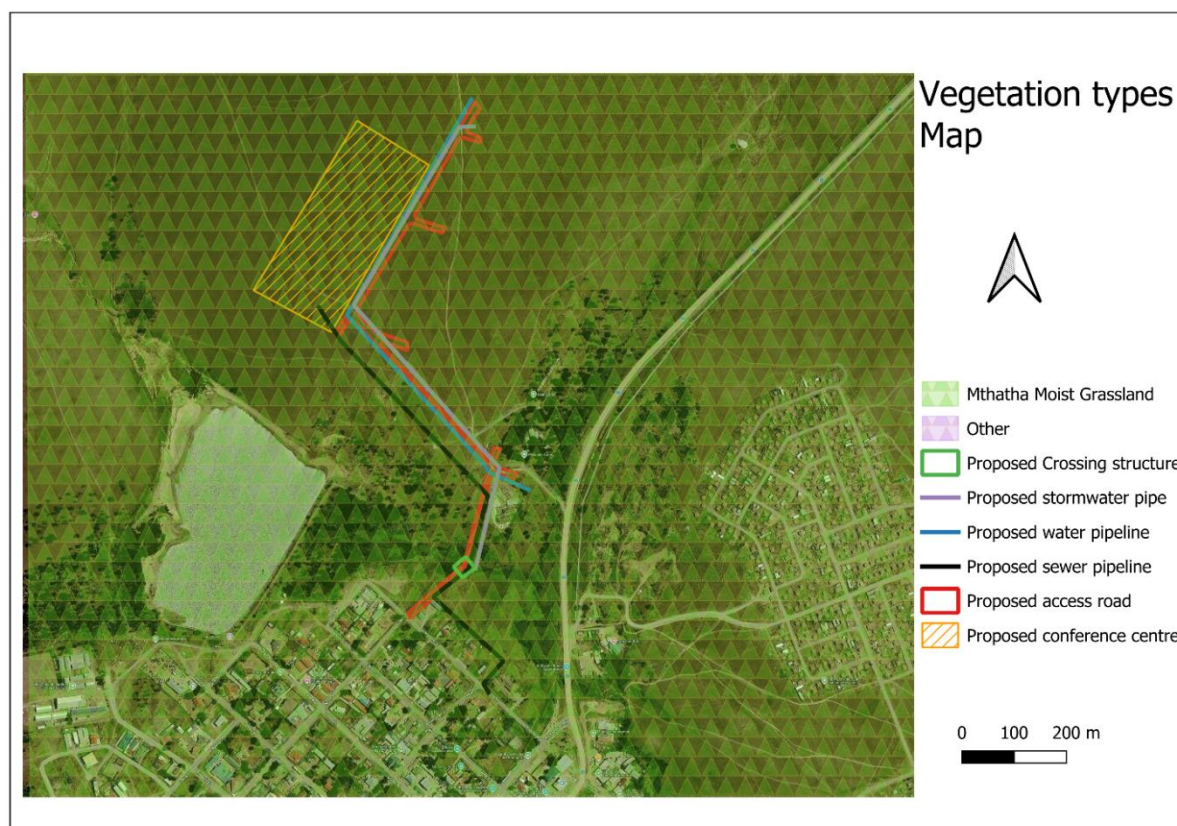


Figure 6: The vegetation map for the proposed development.

Table 6: Common and characteristic plant species of the Mthatha Moist Grassland.

Growth form	Species
Small tree	<i>Acacia natalitia</i>
Graminoid	<i>Cyperus obtusiflorus</i> var. <i>obtusiflorus</i> , <i>Elionurus muticus</i> , <i>Eragrostis curvula</i> , <i>Heteropogon contortus</i> , <i>Microchloa caffra</i> , <i>Paspalum dilatatum</i> , <i>Sporobolus africanus</i> , <i>Themeda triandra</i> , <i>Abildgaardia ovata</i> , <i>Alloteropsis semialata</i> subsp. <i>eckloniana</i> , <i>Aristida congesta</i> , <i>Brachiaria serrata</i> , <i>Chloris virgata</i> , <i>Cymbopogon marginatus</i> , <i>Cynodon dactylon</i> , <i>Cyperus haematocephalus</i> , <i>C. obtusiflorus</i> var. <i>flavissimus</i> , <i>Digitaria eriantha</i> , <i>D. ternata</i> , <i>Eragrostis capensis</i> , <i>E. plana</i> , <i>Eustachys paspaloides</i> , <i>Harpochloa falx</i> , <i>Hemarthria altissima</i> ,

	<i>Hyparrhenia hirta, Panicum ecklonii, Paspalum scrobiculatum, Setaria nigrirostris, Tristachya leucothrix.</i>
Herb	<i>Senecio coronatus, Centella asiatica, Chamaecrista mimosoides, Cyanotis speciosa, Eriosema salignum, Falkia repens, Helichrysum rugulosum, Indigofera hedyantha, I. hilaris var. hilaris, Ipomoea crassipes, Lobelia flaccida, Pentanisia prunelloides subsp. latifolia, Senecio retrorsus, Sonchus dregeanus, Vernonia capensis, V. natalensis, Wahlenbergia stellarioides</i>
Low shrub	<i>Senecio pterophorus, Coddia rudis, Erica caffrorum var. caffrorum, Felicia filifolia subsp. filifolia, Hermannia parviflora.</i>
Herbaceous climber	<i>Rhynchosia totta.</i>
Geophytic Herbs	<i>Boophone disticha, Habenaria dives</i>

4.1.2. REGIONAL CONTEXT

4.1.2.1. EASTERN CAPE BIODIVERSITY CONSERVATION PLAN (ECBCP, 2019)

The Eastern Cape Biodiversity Conservation Plan (ECBCP 2019) is a regional systematic biodiversity conservation plan for the Eastern Cape, gazetted October 2020. It aims to avoid further loss or degradation of biodiversity priority areas and ecological support areas. The plan sets certain development guidelines based on a calculated biodiversity score for different landscapes. Basically, the terrestrial areas covered by the plan are designated as Critical Biodiversity 1, 2, or 3 areas, each with specific development recommendations.

The proposed development site falls within terrestrial CBA 2, because of the Mthatha Moist grassland, which is an Endangered vegetation type. According to the ECBCP, land use that results in vegetation loss in critical biodiversity areas must be avoided. The Mthatha Moist grassland is also currently not protected, but the long-term target is to have 23% of the area protected. Terrestrially, the project area falls under BLMC 2, these areas are near natural

Landscapes. The recommended land use objective for these areas is to maintain biodiversity in a near-natural state with minimal loss of ecosystem integrity. No transformation of natural

habitat should be allowed. In terms of Aquatic CBAs identified by the ECBCP, the appointed aquatic specialist will include them in their assessment.

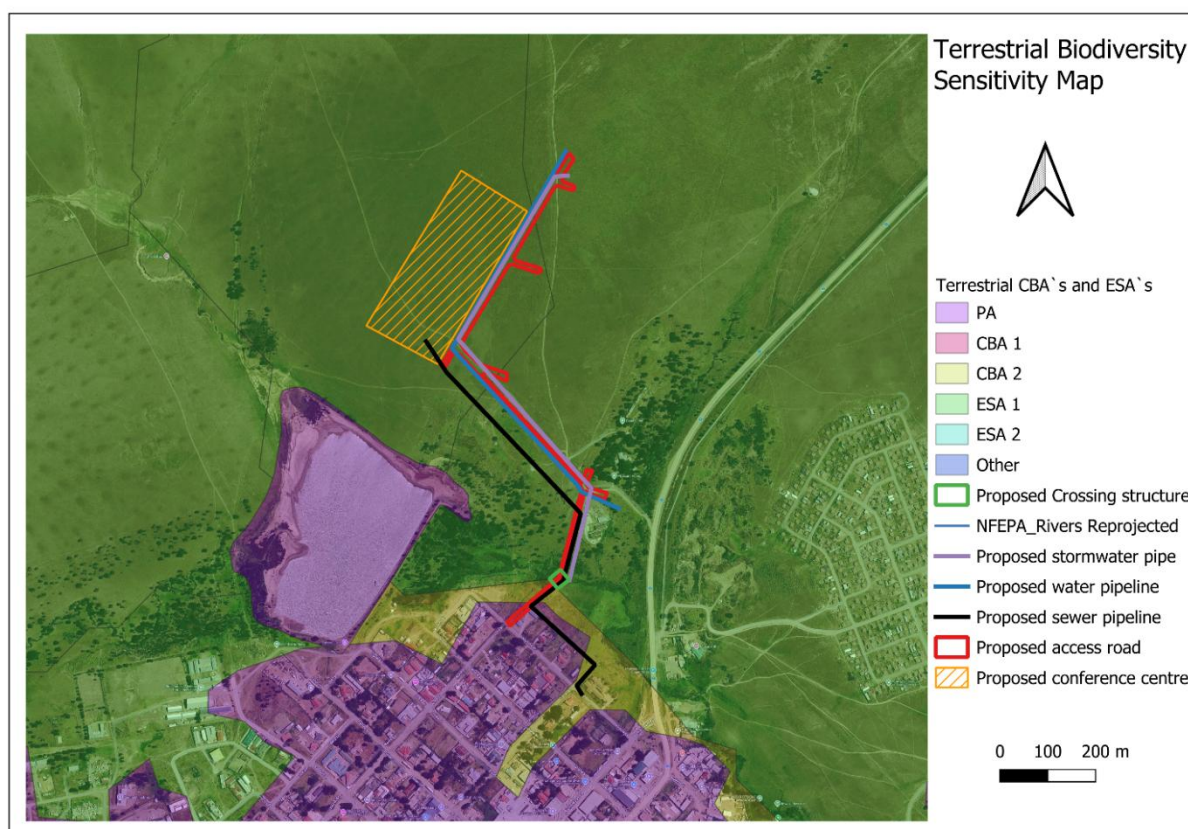


Figure 7: CBAs and ESAs within and surrounding the proposed development site.

4.1.3. LOCAL CONTEXT

Mbashe Greening Strategy

To improve environmental degradation in the municipality, the Mbashe Greening Strategy plans to increase tree cover and promote good environmental governance, especially planning, developing and conservation within the municipal area. Idutywa is identified as a priority town for this greening strategy.

4.2. NATIONAL LIST OF THREATENED ECOSYSTEMS

The NEMBA (Act 10 of 2004) provides for the listing of threatened or protected ecosystems, in one of four categories: Critically Endangered, Endangered, Vulnerable or Protected. Threatened ecosystems are listed to reduce the rate of ecosystem and species extinction by preventing further degradation and loss of structure, function and composition of threatened ecosystems. The purpose of listing protected ecosystems is primarily to conserve sites of exceptionally high conservation value (SANBI, Biodiversity Geographic Information Systems (BGIS)). Importantly, any land-use change application occurring within an ecosystem listed as

Critically Endangered or Endangered in terms of the Biodiversity Act will automatically require environmental authorisation. The proposed development site is located in Mthatha Moist Grassland which is listed as an Endangered ecosystem type in the List of Threatened Ecosystems (2021) and the SANBI's Red list of Ecosystems.

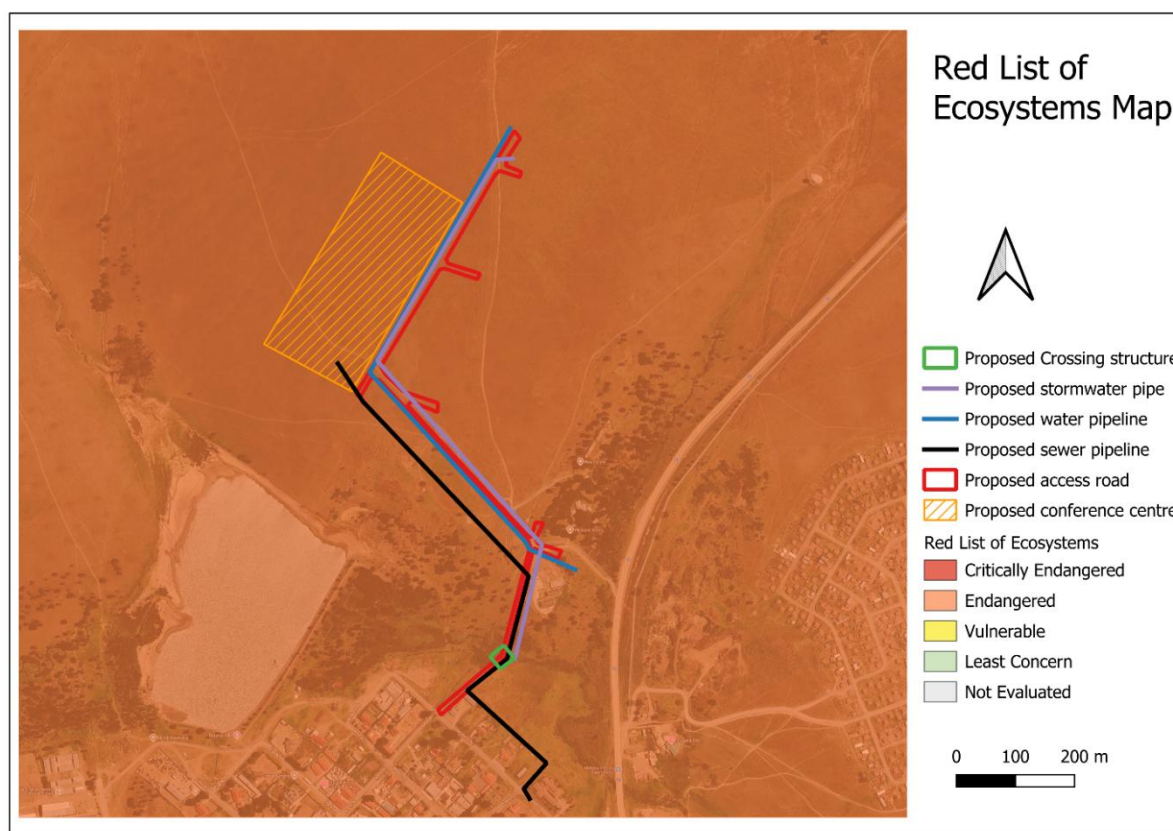


Figure 8: Map showing Red Listed ecosystem types within the proposed development footprint.

4.3. PROTECTED AREAS

Protected areas are areas of land or sea that are protected by law and managed mainly for biodiversity conservation (NPAES, 2016). Protected areas are declared under the National Environmental Management: Protected Areas Act (Act 57 of 2003). The Protected Areas Act provides for several categories of protected areas, including special nature reserves, national parks, nature reserves, marine protected areas and protected environments. Development is regulated within protected areas, as well as buffer areas around them.

The National Protected Area Expansion Strategy (NPAES) presents a 20-year strategy for the expansion of protected areas in South Africa, as they currently do not adequately conserve a representative portion of South Africa's biodiversity (DEA 2016). NPAES identifies priority areas where the expansion of protected areas should take place. Similar priority areas have

been identified on a provisional scale, namely the Eastern Cape Protected Area Expansion Strategy (ECPAES 2010). The Mthatha Moist grassland is currently not protected, but the long-term target is to have 23% of the area protected (NPAES, 2018). The proposed development site does not occur within or near any protected area or its formal buffer zone; therefore, direct physical impacts are not expected.

4.4. STRATEGIC WATER SOURCE AREAS

Strategic Water Source Areas (SWSAs) are defined as areas of land that either:

- (a) supply a disproportionate (i.e. relatively large) quantity of mean annual surface water runoff concerning their size and so are considered nationally important; or
- (b) have high groundwater recharge and where the groundwater forms a nationally important resource; or
- (c) areas that meet both criteria (a) and (b) (Le Maitre *et al.*, 2018).

They include transboundary Water Source Areas that extend into Lesotho and Swaziland. Several river systems in the Eastern Cape, including the Gamtoos, Keiskamma, Mbashe, and Mzimvubu, are fed by upper catchments that experience disproportionately high rainfall and are considered “water factories” of South Africa (ECBCP, 2019). SWSAs are mapped at a national level and represent areas where 50% of South Africa’s rain falls over less than 8% of the land area. Initiatives aimed at managing these SWSAs for enhanced downstream water quality and quantity are underway. Groundwater Strategic Areas with high rates of recharge were identified as well, and cover 9% of SA. The proposed development site falls within the Mzimvubu to Keiskamma water management area, and the Mbashe sub water management area.

4.5. NATIONAL FRESHWATER ECOSYSTEM PRIORITY AREAS

The National Freshwater Ecosystem Priority Areas (NFEPA) project is a collaborative national initiative that aims to conserve South Africa’s freshwater ecosystems and promote the sustainable use of water resources. It does so by identifying Freshwater Ecosystem Priority Areas (FEPAs) and providing strategies to protect them in line with national biodiversity goals

for freshwater systems. These goals are guided by the National Water Act (Act 36 of 1998), the National Environmental Management: Biodiversity Act (Act 10 of 2004), and the National Environmental Management: Protected Areas Act (Act 57 of 2003) ((Nel et al., 2011)0.

The proposed development falls within an upstream management area (these sub catchments must be carefully managed to protect vulnerable downstream river FEPAs and fish support areas from harm) (Figure 17). NFEPA's will be included in the appointed aquatic specialist's assessment.

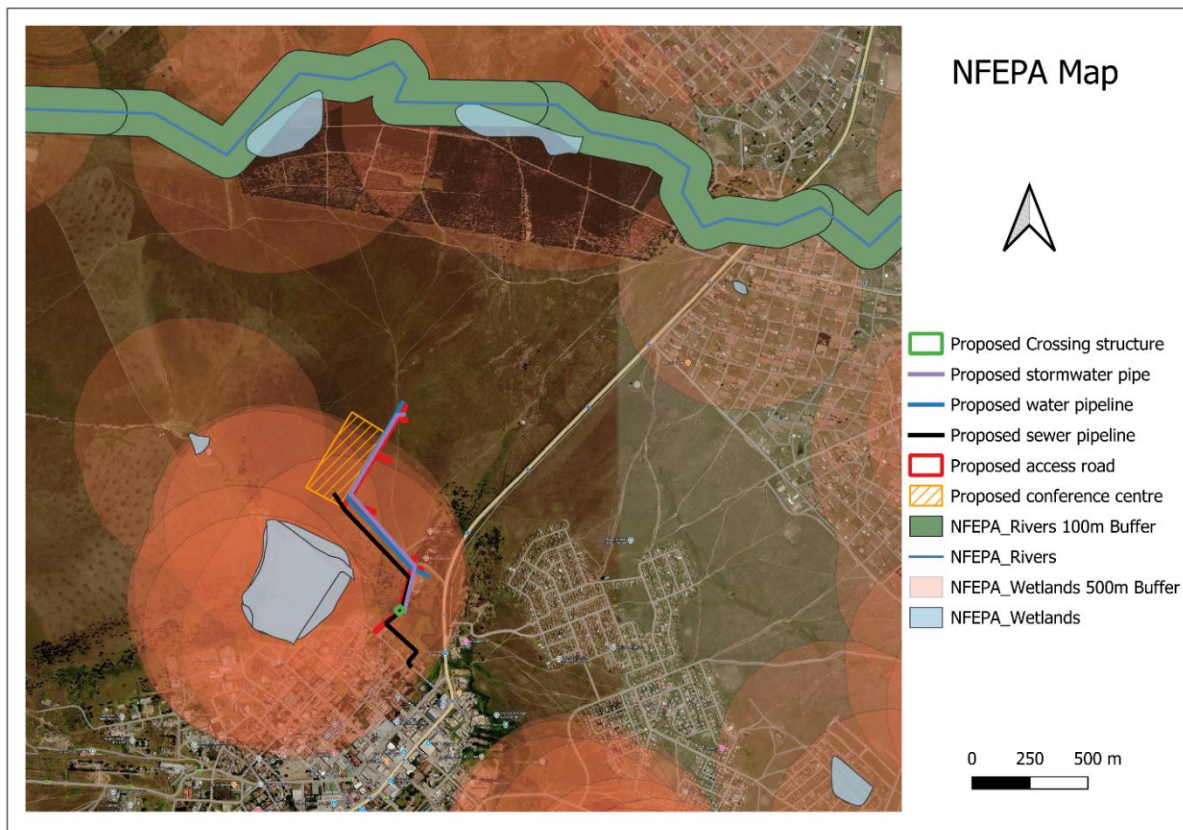


Figure 9: NFEPA around the development site.

4.6. FOREST PATCHES

Forest is protected under the National Forest Act, Act 84 of 1998. A permit is required to disturb the forest. Patches of forest have been mapped at various scales in South Africa. There are no forest patches within the development footprint.

4.7. SPECIES OF CONSERVATION CONCERN

4.7.1. CONSERVATION STATUS

South Africa uses the internationally endorsed IUCN Red List Categories and Criteria in the Red List of South African plants (SANBI, 2020b). This scientific system is designed to measure species' risk of extinction. The purpose of this system is to highlight those species that are most urgently in need of conservation action.

Due to its strong focus on determining the risk of extinction, the IUCN system does not highlight species that are at low risk of extinction. Still, it may nonetheless be of high conservation importance. Because the Red List of South African plants is used widely in South African conservation practices such as systematic conservation planning or protected area expansion, we use an amended system of categories (Table 7: National Red List categories) designed to highlight those species that are at low risk of extinction but of conservation concern.

Table 7: National Red List categories

Category	Acronym	Definition
Extinct	EX	A species is Extinct when there is no reasonable doubt that the last individual has died
Extinct in the Wild	EW	A species is Extinct in the Wild when it is known to survive only in cultivation or as a naturalized population (or populations) well outside the past range
Regionally Extinct	RE	A species is Regionally Extinct when it is extinct within the region assessed (in this case, South Africa), but wild populations can still be found in areas outside the region
Critically Endangered, Possibly Extinct	CR PE	Possibly Extinct is a special tag associated with the category Critically Endangered, indicating species that are highly likely to be extinct, but the exhaustive surveys required for classifying the species as Extinct have not yet been completed. A small chance remains that such species may still be rediscovered
Critically Endangered	CR	A species is Critically Endangered when the best available evidence indicates that it meets at least one of the five IUCN criteria for Critically Endangered, indicating that the species is facing an extremely high risk of extinction
Endangered	EN	A species is Endangered when the best available evidence indicates that it meets at least one of the five IUCN criteria for Endangered, indicating that the species is facing a very high risk of extinction

Category	Acronym	Definition
Vulnerable	VU	A species is Vulnerable when the best available evidence indicates that it meets at least one of the five IUCN criteria for Vulnerable, indicating that the species is facing a high risk of extinction
Near Threatened	NT	A species is Near Threatened when available evidence indicates that it nearly meets any of the IUCN criteria for Vulnerable, and is therefore likely to become at risk of extinction in the near future
Critically Rare	-	A species is Critically Rare when it is known to occur at a single site, but is not exposed to any direct or plausible potential threat and does not otherwise qualify for a category of threat according to one of the five IUCN criteria
Rare	-	A species is Rare when it meets at least one of four South African criteria for rarity, but is not exposed to any direct or plausible potential threat and does not qualify for a category of threat according to one of the four IUCN criteria: • Restricted range: Extent of Occurrence (EOO) <500 km², OR • Habitat specialist: Species is restricted to a specialized microhabitat so that it has a very small Area of Occupancy (AOO), typically smaller than 20 km², OR • Low densities of individuals: Species always occurs as single individuals or very small subpopulations (typically fewer than 50 mature individuals) scattered over a wide area, OR • Small global population: Less than 10 000 mature individuals
Least Concern	LC	A species is Least Concern when it has been evaluated against the IUCN criteria and does not qualify for any of the above categories. Species classified as Least Concern are considered at low risk of extinction. Widespread and abundant species are typically classified in this category
Data Deficient Insufficient Information	- DDD	A species is DDD when there is inadequate information to make an assessment of its risk of extinction, but the species is well defined. Listing of species in this category indicates that more information is required and that future research could show that a threatened classification is appropriate.

Category	Acronym	Definition
Data Deficient - Taxonomically Problematic	DDT	A species is DDT when taxonomic problems hinder the distribution range and habitat from being well defined, so that an assessment of risk of extinction is not possible.
Not Evaluated	NE	A species is Not Evaluated when it has not been evaluated against the criteria. The national Red List of South African plants is a comprehensive assessment of all South African indigenous plants, and therefore all species are assessed and given a national Red List status. However, some species included in Plants of southern Africa: an online checklist are species that do not qualify for national listing because they are naturalized exotics, hybrids (natural or cultivated), or synonyms. These species are given the status Not Evaluated and the reasons why they have not been assessed are included in the assessment justification.

Threatened species are species that are facing a high risk of extinction. Any species classified in the IUCN categories Critically Endangered, Endangered or Vulnerable is a threatened species (Figure 10: Threatened species categories (SANBI, 2020a).). Species of conservation concern (SCC) are species that have a high conservation importance in terms of preserving South Africa's high floristic diversity and include not only threatened species, but also those classified in the categories Extinct in the Wild (EW), Regionally Extinct (RE), Near Threatened (NT), Critically Rare, Rare, Declining and Data Deficient - Insufficient Information (DDD). All South African plant species have been rated, according to their extinction threat, using criteria adapted by SANBI.

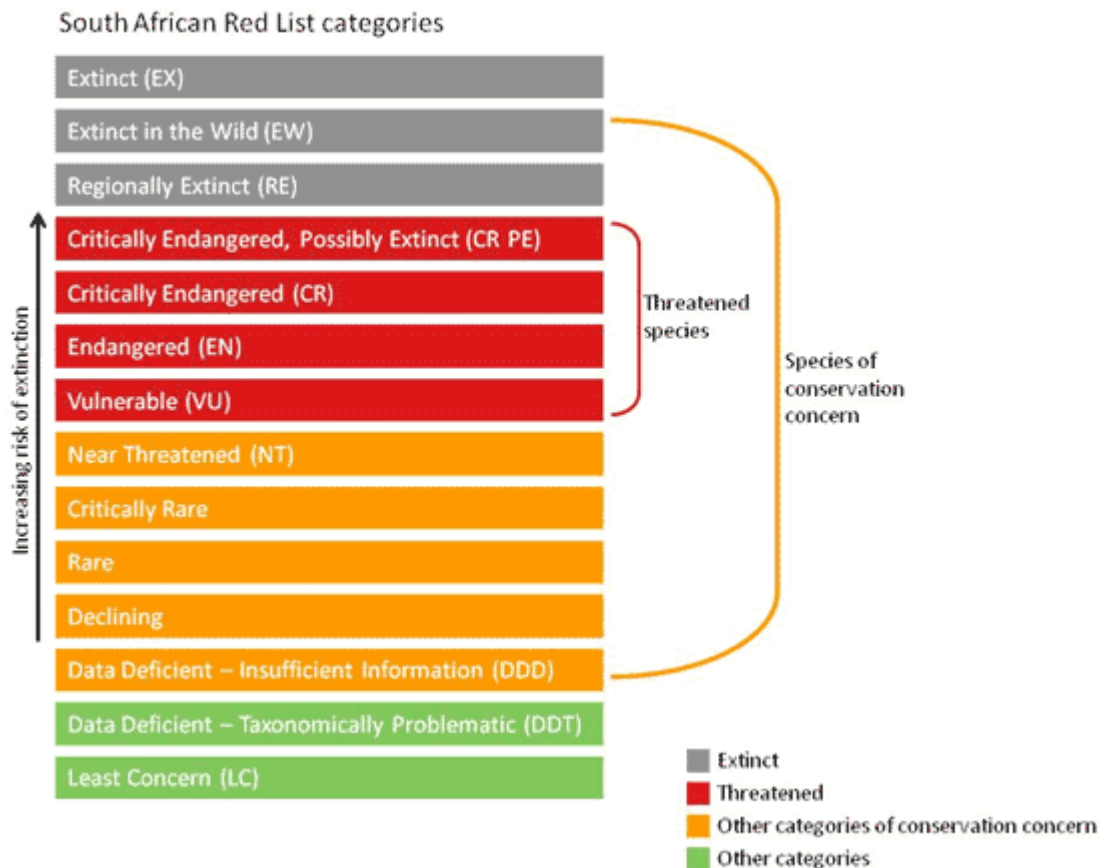


Figure 10: Threatened species categories (SANBI, 2020a).

4.7.2. PLANT SPECIES

Several data sources exist to identify the location of SCCs. The Threatened Species Programme (TSP) has a database of locations of SCCs. This is based on herbarium records, as well as recent searches by experts, and volunteer programmes such as the Custodians of Rare and Endangered Wildflowers (CREW). iNaturalist, a citizen science electronic platform for posting pictures of species to be identified by experts, is also a source.

The Probability of Occurrence (PoO) is rated according to the availability of suitable habitat (Table 8), and whether the species has been previously recorded in the general vicinity of the assessed area. Data for habitat requirements was based on the species descriptions of the Red List (SANBI, 2020b). The Probability of Occurrence was rated as:

Table 8: Criteria for Determining the Probability of Species Occurrence.

Probability of Occurrence	Description
Low	No suitable habitat or confirmed records of the species occur within the proposed area of influence (PAOI), and targeted searches for easily identifiable species yielded no observations.
Medium	Unclear/unknown habitat requirements are but species recorded within 20 kms of PAOI; or degraded habitat occurs within PAOI; or habitat occurs within PAOI but no identification records within the vicinity of the PAOI
High	Suitable habitat occurs within the PAOI, and the species has been recorded nearby.
Confirmed	Species was identified within the PAOI

The web-based Screening Tool identified five plant Species of Conservation Concern (SCCs) with a medium sensitivity rating, indicating that the site may contain modelled habitat for threatened species. However, according to the NEWPOSA, there are no confirmed (recorded) or predicted (modelled) SCC occurrences within the proposed development footprint, beyond those flagged by the Screening Tool. This suggests that, apart from the Screening Tool predictions, there is no additional evidence of SCCs expected or known to occur in the area.

4.8. PREDICTED AREA OF IMPACT

The Predicted Area of Influence (PAOI) is defined according to important ecosystem processes and functions that may be plausibly affected by the proposed development and its associated activities.

5. ECOLOGICAL ASSESSMENT

5.1. VEGETATION COMMUNITY COMPOSITION

The proposed conference centre development is situated in an undeveloped area north-east of the Dutywa town and its suburban area. It is accessed via a gravel road branching off from the N2. The proposed access road is located to the south-east of the proposed conference centre footprint. Cattle dung and sheep/goat droppings were observed on site, negatively impacting the vegetation. Much of the vegetation on site can be considered secondary grassland. The dominant vegetation type on site is Mthatha Moist Grassland, in agreement with VegMap (SANBI, 2018). It is mainly grass species interspersed with forb species. The grass layer was dominated by *Themeda triandra*, *Cymbopogon* sp, *Eragrostis* sp. *Sporobolus* sp. and *Heteropogon contortus*. The dominant forbs were *Helichrysum pallidum*, *Helichrysum herbaceum*, *Oxalis obliquifolia*, and *Gerbera piloselloides*. There was also *Vachellia* sp. There weren't any trees/ shrubs in the proposed conference centre footprint, but there were in the trees and shrubs near the proposed access road site. In general, the area is relatively degraded and species-poor. There was an informal pathway and little litter. These suggest ongoing human use of the area. Although the vegetation on site is secondary grassland and has invasive plants, it can be considered largely intact and functional. There was a lot of *Solanum linnaeanum*, which indicates disturbance, possibly overgrazing.



Figure 11: Site conditions and vegetation composition showing dominant grasses and forbs.

5.2. SPECIES OF CONSERVATION CONCERN

No plant Species of Conservation Concern (SCC) were identified on site and by the Screening Tool. In addition, the NEWPOSA database was consulted, which indicated that no SCCs are expected to occur within the project area. All the plant species identified on site are listed as Least Concern.

5.3. THREATENED OR PROTECTED SPECIES

The following legislation was consulted to determine whether a species is protected by Legislation:

- National Environmental Management: Biodiversity Act 10 of 2004 – Regulations pertaining to Threatened or Protected Terrestrial Species and Freshwater Species (GNR 3009 of 2023);
- Nature and Environmental Conservation Ordinance of 1974; and
- National Forests Act No. 84 of 1998 – List of Protected Trees (published 8 September 2017).

The plant species recorded on site where: *Themeda triandra*, *Cymbopogon* sp, *Eragrostis* sp. *Sporobolus* sp, *Heteropogon contortus*, *Helichrysum pallidum*, *Helichrysum herbaceum*, *Oxalis obliquifolia*, and *Gerbera piloselloides* and *Vachellia* sp. All of these are classified as Least Concern according to the SANBI Red List. However, while there were no plants that were threatened, the site falls within the Endangered Mthatha Moist Grassland, which is a threatened vegetation type.

5.4. ALIEN INVASIVE PLANTS

A landowner must remove or manage all Invasive Alien Species (IAS) on their property. AIS are defined by the National Environmental Management: Biodiversity Act, Act 10 of 2004: Alien and Invasive Species Lists of 2020. Three alien species were identified on site, and both are listed as Alien Invasive Species (AIS). A list of AIS on-site is included in Table 9. All Category 1a and 1b species must be controlled and removed from the property. A permit is required to keep Category 2 species. Category 3 species may remain but must be controlled and no further planting is allowed.



Table 9: List of invasive alien species (IAS) on site.

Species	Category
<i>Acacia mearnsii</i> De Wild.	2
<i>Solanum mauritianum</i> Scop.	1b
<i>Carduus nutans</i>	1b

5.5. ECOLOGICAL SENSITIVITY AND SITE ECOLOGICAL IMPORTANCE

The sensitivity map was derived by identifying the conservation and biodiversity priorities of the site and ground-truthing them during a site visit. The desktop assessment considered the following conservation tools and plans:

- Online Screening Tool Report;
- Provincial Biodiversity Plans (CBAs, ESAs, PAs); and
- NEMBA Threatened Ecosystems.

According to a desktop assessment of the site's ecological sensitivity, using datasets identified in the Terrestrial Biodiversity Impact Assessment Protocols (2020), and the site visit, the proposed development has a high ecological sensitivity. The proposed development occurs within Endangered Mthatha Mositi grassland, CBA 2 and ESA 1, it also falls within the Mzimvubu to Keiskamma Water Management Area, the Mbashe Sub-Water Management Area, and an upstream FEPA management area. A wetland also occurs adjacent to the proposed development area, with the mapped wetland buffer extending into the proposed development footprint. These features make the ecological sensitivity of the site high, since the area maintains hydrological and ecological processes.

The proposed conference centre is situated within a large area of intact grass vegetation. Although the grassland is secondary, it doesn't have bare patches or visible signs of erosion and therefore continues to provide ecological functions and connectivity to the surrounding vegetation. While there is a considerable amount of weedy, *Solanum linnaeanum*, possibly due to overgrazing, the conservation of the remaining grassland habitat is important for maintaining ecosystem integrity. Therefore, this area has a high sensitivity.

While the proposed access road and associated pipelines occur in a more disturbed area than the proposed conference centre, particularly where they approach the existing residential area, the access road also crosses a tributary of the Ngxakaxa River. Construction in this section has the potential to directly disturb the watercourse and associated habitat, as well as alter local hydrological processes. Field observations recorded increased litter and alien invasive species, including bugweed and black wattle, in this corridor. Given these factors, the ecological sensitivity of this section is considered medium.

Table 10: Evaluation of SEI of vegetation communities and other habitats in the PAOI for the proposed Dutywa conference centre and associated infrastructure per SANBI (2020). (BI = biodiversity importance, RR = receptor resilience).

Infrastructure	Area (ha)	Description	SEI
Proposed conference centre	6.18	BI = CI + FI CI = Low because the no SCC's were flagged by the screening tool, and seen on site, and the development footprint is small and the development is in secondary grassland not natural grassland. FI= High, because the footprint is in the middle of continuous intact grass layer and there was no visible erosion, no significant bare patches, and little litter. The area is connected to the surrounding grassland BI = Low + High = Medium	SEI = BI + RR SEI=Medium + Low = High
Proposed access road	1.45	BI = CI + FI CI = Low because the no SCC's were flagged by the screening tool, and seen on site, and the development footprint is small and the development is in secondary grassland not natural grassland.	SEI = BI + RR SEI= Low + Low = Medium

Infrastructure	Area (ha)	Description	SEI
		FI= medium because only a portion of the proposed access road there is part of the grassland, the other section is closer to a residential area and more disturbance, e.g Litter and alien invasive species, including Bugweed and Black Wattle BI= Low + Medium= Low	
Total area	7.63		

6. IMPACT ASSESSMENT

6.1. POTENTIAL IMPACTS

The largest potential impact of the proposed development is the loss of natural vegetation. The potential impact will mostly be limited to the footprint of the conference centre and associated infrastructure. Recommendations are provided below to mitigate and/or minimise the impact of these issues in sensitive areas mapped within the development footprint.

6.1.1. CONSTRUCTION PHASE IMPACTS

6.1.1.1. DIRECT IMPACTS

- i. Loss of indigenous vegetation

The proposed development will result in the loss of portions of the Mthatha Mosit grassland and associated habitat. Construction activities will also lead to topsoil removal and compaction, with associated risks of altered surface hydrology, soil disturbance, and increased erosion. Disturbance from noise and vibration is expected to affect both livestock and the

nearby community. The affected vegetation units include the Mthatha moist grassland, which is classified as an Endangered vegetation type.

Table 11: Potential impacts of the construction phase - loss of vegetation.

Direct loss of indigenous vegetation				
	PHASE		CONSTRUCTION	
	TYPE		NEGATIVE, DIRECT, CUMULATIVE	
	SIGNIFICANCE		WITH MITIGATION	
EXTENT	1	Medium (52)	1	Medium (44)
DURATION	4		4	
MAGNITUDE	8		6	
PROBABILITY	4		4	
Irreplaceable Loss	The resource will be lost			
Reversibility	Irreversible			
Impact Significance	Medium			
MITIGATION:	1. Minimise natural vegetation clearance: Limit the footprint of disturbance by maximising the use of existing tracks, roads, and infrastructure corridors wherever possible. 2. Environmental supervision: Appoint a competent Environmental Control Officer (ECO) for the construction phase to ensure all recommendations in this report are implemented. 3. Restrict access: Limit all construction and maintenance access to demarcated areas to prevent disturbance to vegetation outside the development footprint. 4. Protect natural resources: Prohibit collection of plants, wood, or other natural resources outside the approved development footprint. 5. Laydown areas: Restrict laydown, storage, and temporary work areas to sites of low ecological sensitivity to reduce unnecessary habitat loss. 6. Selective clearing: Only clear vegetation that is essential for construction, such as for pipelines and avoid unnecessary widening of access routes, laydown areas, material storage areas and construction camps. 7. Topsoil management: Remove and stockpile topsoil carefully during construction for use in the rehabilitation of disturbed areas. 8. Rehabilitation timing: Undertake rehabilitation immediately after construction activities cease in a given area to reduce erosion and promote ecosystem recovery. 9. Rehabilitation methods: Allow mostly passive regeneration of vegetation, supplemented initially with indigenous grass mixes to stabilise soils and accelerate recovery.			

	Protected species management: Apply for relevant permits prior to removing any protected species, and ensure removal is conducted in accordance with legal requirements.
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No plant SCCs were flagged by the Screening Tool for the assessment area, and no plant SCCs were recorded during the site visit. Based on the observed habitat condition and field and desktop assessment, no SCCs are expected to occur within the proposed development footprints. Consequently, no direct loss of plant SCCs is expected as a result of the proposed development.

ii. Loss of Ecosystem Function and Connectivity

The proposed project will significantly affect the ecological function and connectivity of the conference centre footprint. Although the footprint lies within secondary grassland, it supports a continuous grass layer with good cover and shows no major bare patches or erosion. Clearing this vegetation will disrupt ecosystem services, including increased runoff and sedimentation into the nearby river, and reduce habitat connectivity within the broader grassland landscape. Fragmentation will interfere with natural ecological processes such as seed dispersal, pollination, and the spread of fire, the latter being a critical driver in maintaining grassland ecosystems. Grazing opportunities will also be reduced, potentially increasing pressure on surrounding grassland areas.

The proposed access road and pipelines will add further disturbance by creating a linear break through the grassland, reducing habitat connectivity and contributing to edge effects and fragmentation. Together, these impacts will diminish ecological functioning and resilience across the affected footprint.

Table 12: Potential impact of loss of ecosystem functioning.

Loss of Ecosystem function and connectivity		
	PHASE	CONSTRUCTION
	TYPE	NEGATIVE, DIRECT
	SIGNIFICANCE	WITH MITIGATION

EXTENT	2	Medium (36)	1	Low (24)
DURATION	4		3	
INTENSITY	6		4	
PROBABILITY	4		3	
Irreplaceable Loss	Resource will be partly lost			
Reversibility	Moderately reversible			
Mitigation Potential	Moderate			
MITIGATION:	1. Minimise natural vegetation clearance as far as possible by: minimising the development footprint; avoiding unnecessary clearing or disturbance of surrounding grassland; and avoiding unnecessary widening of the road ; 2. Secondary and Tertiary PAOI are no-go areas, and no development activities are allowed.			

iii. Disturbance of the surface resulting in increased risk of AIPs

Clearing for the construction phase of the project will result in soil disturbance and reduced cover of indigenous vegetation, greatly increasing the chance of the establishment of alien invasive plants (Table 10). Several species, including *Acacia mearnsii*, *Solanum mauritianum*, and *Carduus nutans*, are likely to invade the disturbed areas.

Table 13: Potential risk of spread and invasion of AIPs

IMPACT	Natural vegetation is negatively affected by the increased risk of invasion by AIPs due to the construction of the proposed development				
NATURE	NEGATIVE		TYPE	DIRECT, CUMULATIVE	
PHASE		SIGNIFICANCE		WITH MITIGATION	
Construction	EXTENT	2	Medium (48)	2	Insignificant (8)
	DURATION	2		1	
	INTENSITY	8		1	
	PROBABILITY	4		2	
	Irreplaceable Loss	Moderate loss			
	Reversibility	High reversibility			
	Mitigation Potential	High			

	MITIGATION:	1. Monitor alien invasive plant species during construction and implement routine removal of the species 2. Where required, rehabilitate and revegetate areas as soon as possible using indigenous plant species
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6.1.2. OPERATIONAL PHASE IMPACTS

6.1.2.1. DIRECT IMPACTS

i. Direct Increase in Anthropogenic Disturbance

The operation of the development will result in increased anthropogenic influences on the surrounding natural vegetation. These impacts may manifest as edge effects, including increased disturbance, noise, littering, and plant collection, which could lead to further loss or disruption of species and ecosystem processes on site. Other disruptive activities such as dumping, clearing of natural vegetation, overgrazing, and the creation of uncontrolled footpaths, may also increase the risk of invasion by Invasive Alien Plants.

Table 14: Potential risk of increased anthropogenic disturbances

IMPACT	Direct Increase in Anthropogenic Disturbance during operation of the developed area				
NATURE	NEGATIVE		TYPE	DIRECT, CUMULATIVE	
PHASE		SIGNIFICANCE		WITH MITIGATION	
Operation	EXTENT	2	Low (30)	2	Low (20)
	DURATION	2		1	
	INTENSITY	6		2	
	PROBABILITY	3		2	
	Irreplaceable Loss	Moderate loss			
	Reversibility	High reversibility			
	Mitigation Potential	High			

	MITIGATION:	1. Implement the Open Space Maintenance Plan 2. Limit the use of pesticides and herbicides during the operation phase, particularly with the control of AIS; 3. Access should be limited to prevent further disturbance to the vegetation outside of the development footprint; 4. No fires should be allowed on site; 5. No plant or wood collection should be allowed outside of the proposed development footprint;
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6.1.3. CUMULATIVE IMPACTS

The proposed conference centre is located within secondary grassland associated with the Endangered Mthatha Moist Grassland ecosystem type. The site is subject to existing anthropogenic pressures, particularly livestock grazing, with evidence of disturbance including the presence of weedy species and some localised litter. Despite these pressures, the conference centre footprint retains a relatively continuous grass layer and vegetation cover, with no extensive bare areas or visible soil erosion observed during the field assessment. The grassland therefore retains ecological functioning and contributes to connectivity with the surrounding grassland landscape.

The operation of the proposed conference centre is likely to introduce additional anthropogenic pressures through increased human activity, vehicular movement, artificial lighting, waste generation and ongoing maintenance. These activities may result in continued disturbance of surrounding vegetation, increased opportunities for the establishment and spread of alien and invasive plant species, and disturbance or displacement of fauna and livestock. The permanent transformation of approximately 6.18 ha of secondary grassland will also result in a localised reduction in habitat extent and ecosystem functioning, while the associated access road and infrastructure may contribute to further fragmentation and reduction in connectivity.

In combination with existing land-use pressures and surrounding development, the proposed conference centre may contribute cumulatively to the incremental loss and fragmentation of remaining grassland within the landscape. Although the proposed footprint represents a

relatively small portion of the broader Mthatha Moist Grassland ecosystem, continued transformation of remaining grassland may progressively reduce habitat availability, ecological connectivity and ecosystem functioning. The potential cumulative impacts can be reduced through minimising the permanent development footprint, retaining surrounding grassland and ecological linkages, preventing unnecessary disturbance, controlling alien invasive species, appropriately managing stormwater and waste, and rehabilitating areas temporarily disturbed during construction.

Table 15: Potential impact of cumulative loss of natural vegetation.

IMPACT	Cumulative loss of natural vegetation due to clearing and habitat degradation				
NATURE	NEGATIVE		TYPE	DIRECT, CUMULATVE	
PHASE		SIGNIFICANCE		WITH MITIGATION	
Construction Operation	EXTENT	2	Medium (33)	2	Low (18)
	DURATION	3		3	
	INTENSITY	6		4	
	PROBABILITY	3		3	
	Irreplaceable Loss	Moderate loss			
	Reversibility	Low reversibility			
	Mitigation Potential	Moderate			
	MITIGATION:	1. Prevent development from occurring within the identified No Go Areas (i.e surrounding vegetation).			

6.1.4. NO-GO OPTION

Under the No-Go Option, the proposed conference centre and associated infrastructure would not be developed, and the site would remain in its current condition. In the absence of the proposed development, the existing grassland would remain available for livestock grazing and would continue to provide habitat and ecological connectivity within the surrounding landscape. Existing pressures, including grazing and the occurrence of weedy and alien invasive species, are likely to persist. However, the permanent transformation of secondary grassland associated with the conference centre would be avoided, as would the additional disturbance associated with the proposed access road and associated infrastructure.

The No-Go Option would therefore avoid the direct loss and permanent transformation of grassland, as well as potential reductions in ecosystem function and connectivity associated with the proposed development. It would also avoid additional operational pressures associated with increased human activity, vehicular movement, waste generation, artificial lighting and stormwater runoff. The wetland, tributary and associated ecological processes would remain undisturbed by the proposed development. However, the No-Go Option would not address existing pressures on the site, including livestock grazing and alien invasive species, and these would likely continue under the current land-use regime.

7. IMPACT STATEMENT

An impact statement is required in accordance with the NEMA EIA Regulations for the proposed conference centre and associated infrastructure. No feasible development alternatives were assessed as part of this terrestrial biodiversity assessment.

According to the National Screening Tool, the Terrestrial Biodiversity Theme for the proposed development area is rated as Very High sensitivity, associated with the occurrence of the Endangered Mthatha Moist Grassland ecosystem type and mapped CBA 2. Portions of the proposed development footprint also overlap ESA 1. The site also occurs within an FEPA-related upstream management area. Field verification confirmed that the proposed development footprint comprises secondary grassland associated with the Mthatha Moist Grassland vegetation type. Although the grassland has been subject to existing anthropogenic pressures, including livestock grazing and the occurrence of weedy and alien invasive species, the conference-centre footprint retains a relatively continuous grass layer, good vegetation cover and ecological connectivity with the surrounding grassland. No extensive bare areas or visible soil erosion were observed within the conference-centre footprint.

No plant species of conservation concern were identified by the Screening Tool or recorded during the field assessment. The Conservation Importance of the affected secondary grassland is therefore assessed as Low. However, the habitat retains ecological functionality and connectivity, with the functional integrity of the conference-centre footprint assessed as High and that of the access-road corridor as Medium. The resilience of the affected grassland to major disturbance is considered Low, as substantial vegetation and soil disturbance

associated with development may result in prolonged and potentially incomplete recovery of grassland composition and ecological functionality.

The proposed development will result in the permanent transformation of secondary grassland within the conference-centre footprint, in addition to vegetation clearance and disturbance associated with the access road and other supporting infrastructure. The proposed access road will also cross a tributary and occurs in proximity to a wetland and associated buffer, increasing the potential for impacts on ecological connectivity, freshwater-associated processes and surrounding habitat. The development may result in the loss or reduction of ecosystem function and habitat connectivity, displacement of livestock grazing, increased disturbance associated with human activity and vehicle movement, and increased opportunities for alien invasive plant establishment. Operational stormwater runoff, waste and other activities may additionally affect surrounding terrestrial and aquatic environments if not appropriately managed.

The significance of the residual impacts will depend on strict implementation of the mitigation measures proposed in this assessment and the EMPr. Key measures include minimising the development footprint and unnecessary vegetation clearance; retaining surrounding grassland and ecological linkages; minimising fragmentation associated with the access road; avoiding and minimising disturbance to the wetland, tributary and associated buffer; appropriately managing stormwater and waste; controlling alien invasive species; and rehabilitating all areas temporarily disturbed during construction using appropriate indigenous grassland species and conserved topsoil.

Based on the findings of the terrestrial biodiversity assessment, the proposed development will result in permanent and localised impacts on secondary grassland and associated ecological functioning. Although no SCCs were identified and the affected vegetation is secondary rather than natural grassland, the occurrence of the development within an Endangered ecosystem, CBA 2 and portions of ESA 1 warrants the implementation of stringent avoidance and minimisation measures. The development should therefore be assessed based on the residual impact significance determined following implementation of the recommended mitigation measures, rather than solely on the absence of SCCs.

8. CONCLUSION

The construction of the proposed development will result in the direct loss and disturbance of portions of secondary grassland associated with the Endangered Mthatha Moist Grassland vegetation type. Although the vegetation within the development footprint is subject to existing anthropogenic pressures, including livestock grazing, weedy and alien vegetation, and localised disturbance, it retains ecological functioning and connectivity with the surrounding grassland. Given the ecological characteristics of the site and its occurrence within an Endangered ecosystem, CBA 2 and portions of ESA 1, the overall terrestrial sensitivity of the development footprint is considered High.

Provided that the development remains strictly within the approved footprint and that the recommended avoidance, minimisation, rehabilitation and alien invasive vegetation management measures are fully implemented, the residual impacts on terrestrial biodiversity are expected to be limited and manageable.

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10. APPENDICES